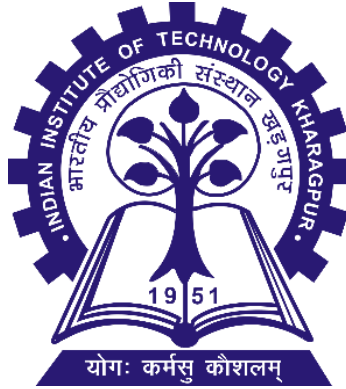




Indian Institute of Technology Kharagpur

Master's Thesis Project II

Infrastructure Finance:

**Volatility Spillovers, Factor Structure, and Portfolio
Optimization of Indian Infrastructure Assets**

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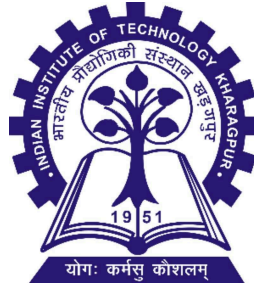
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*A thesis submitted in fulfilment of the requirements
for the degree of Master of Technology*

in

Financial Engineering

**VINOD GUPTA SCHOOL OF MANAGEMENT, INDIAN INSTITUTE OF TECHNOLOGY
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CERTIFICATE

This is to certify that the project report entitled "Infrastructure Finance: Volatility Spillovers, Factor Structure, and Portfolio Optimization of Indian Infrastructure Assets" submitted by Aakansh Singh (Roll No. 21CE3FP20) to Indian Institute of Technology Kharagpur towards partial fulfillment of requirements for the award of degree of Master of Technology in Financial Engineering is a record of bona fide work carried out by him under my supervision and guidance during Spring Semester, 2025-26.

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Declaration

I, Aakash Singh (Roll No. 21CE3FP20), hereby declare that the report entitled

“Infrastructure Finance: Volatility Spillovers, Factor Structure, and Portfolio Optimization of Indian Infrastructure Assets”

submitted in partial fulfilment of the requirements for the award of the degree of Master of Financial Engineering at Indian Institute of Technology Kharagpur, is an original work carried out by me under the supervision of Dr. Rudra Prakash Pradhan, Vinod Gupta School of Management, IIT Kharagpur.

I further declare that:

1. This dissertation has not been submitted, in part or in full, to any other university or institute for the award of any degree, diploma or any other academic qualification.
2. The work reported here is the result of my own investigation, except where due acknowledgement has been made.
3. All sources of information used in this dissertation have been duly cited and referenced.

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Abstract

This dissertation provides the first systematic empirical baseline for Indian Infrastructure Investment Trusts (InvITs) and Real Estate Investment Trusts (REITs), a ₹1.5 lakh crore listed asset class that has had limited academic analysis since its introduction in 2017–2019. Using 1,023 daily observations from July 2020 to March 2026 across five category portfolios, the study follows a four-phase approach. It combines GJR-GARCH and DCC-GARCH for volatility modelling, Diebold-Yilmaz (2012) generalised forecast error variance decomposition for spillover analysis, OLS with HAC standard errors and Shapley R^2 for factor identification, Bayesian Ridge regression with hierarchical priors for prediction, and walk-forward portfolio optimisation across eight strategies.

Five main findings emerge. First, Indian InvITs and REITs are strongly segmented from global markets, with only 0.34% and 1.28% of their variance explained by foreign shocks, far below developed-market REITs. Second, a clear three-level factor structure is observed: infrastructure stocks are strongly driven by market factors ($R^2 = 67.8\%$), REITs are weakly explained (6.5%), and InvITs are largely idiosyncratic (8.1%). Third, the Bayesian Ridge model with hierarchical priors achieves strong predictive performance, with an out-of-sample Information Coefficient of 0.254, well above standard benchmarks. Fourth, portfolio optimization using raw predictions delivers a Sharpe ratio of 1.87 before costs and 1.47 after costs, with a 97.7% probability of outperforming the equal-weight benchmark. Fifth, removing Grinold-Kahn scaling improves performance significantly, with a loss of 0.67 Sharpe points observed when it is applied in this setting.

The study contributes a clear empirical baseline for Indian InvITs and REITs, introduces a hierarchical Bayesian framework that can be applied to other emerging asset classes, and provides evidence on when standard signal-scaling methods may reduce performance.

Keywords: Infrastructure Investment Trusts; Real Estate Investment Trusts; market segmentation; volatility spillovers; Diebold-Yilmaz; GJR-GARCH; DCC-GARCH; Bayesian Ridge regression; hierarchical priors; portfolio optimisation; Grinold-Kahn; India.

1. Introduction

Infrastructure is increasingly recognised as a distinct asset class, offering stable cash flows, long-duration contracts, and potential diversification benefits relative to traditional equities. In developed markets, listed infrastructure vehicles such as real estate investment trusts have evolved over time from illiquid, niche instruments into securities that behave much like equities. This transition reflects a broader process in which real asset markets move from segmentation toward integration as liquidity deepens and investor participation expands.

India represents a much earlier stage of this evolution. The introduction of Infrastructure Investment Trusts and REITs over the past decade has created a new channel for infrastructure financing. These vehicles allow investors to access income-generating assets such as toll roads, transmission networks, and commercial real estate through tradable securities. Despite this structural similarity to global counterparts, their economic characteristics may differ substantially. InvITs in particular are backed by operating infrastructure assets with regulated or quasi-fixed cash flows, while REITs are exposed to property market dynamics. Infrastructure-related equities, in contrast, behave more like traditional stocks with direct exposure to economic activity.

The empirical behaviour of these instruments in India remains insufficiently understood. Most existing evidence on infrastructure and real estate securities comes from developed markets, where integration with global financial systems is relatively high. Emerging markets may not follow the same pattern. Lower liquidity, thinner trading, and institutional constraints can lead to segmentation, delayed price discovery, and a greater role for idiosyncratic or non-macro drivers of returns. Whether Indian infrastructure assets exhibit these characteristics, or whether they already behave like their developed-market counterparts, is an open question.

This uncertainty has direct implications for both theory and practice. From a theoretical perspective, the classification of infrastructure assets depends on whether their returns are driven by global factors, domestic macroeconomic variables, or asset-specific characteristics such as liquidity. From a practical perspective, investors need to know whether these assets provide diversification benefits, whether their risks are systematic or idiosyncratic, and whether their returns can be forecasted and exploited in portfolio construction.

A related issue concerns the applicability of standard asset pricing frameworks. Models based on market and macroeconomic factors assume that returns are largely explained by systematic exposures. However, if Indian infrastructure markets are still in an early stage of development, these models may have limited explanatory power. In such cases, alternative mechanisms such as illiquidity, delayed adjustment to information, or institutional frictions may play a more important role.

Even if traditional models struggle to explain returns, this does not preclude the existence of predictive signals. Recent advances in machine learning have shown that combining economic structure with statistical learning can uncover patterns that are not captured by linear models

alone. Whether such approaches are effective in a setting characterised by small sample sizes and low liquidity remains an empirical question. Importantly, the value of any predictive model ultimately depends on whether it can improve investment outcomes once translated into portfolio decisions.

This dissertation examines Indian infrastructure asset classes across four dimensions: their integration with global markets, the drivers of their returns, the predictability of those returns, and the economic value of predictive signals. By analysing these dimensions in sequence, the study aims to provide a coherent understanding of how these assets behave and how they can be used in practice.

1.2 Scope and Approach

The analysis focuses on three categories of Indian infrastructure assets: InvITs, REITs, and infrastructure-related equities. Each category is represented by an equal-weighted portfolio constructed from listed securities. The use of category-level portfolios serves two purposes. It reduces the influence of idiosyncratic shocks associated with individual assets and allows the analysis to focus on systematic patterns that define each asset class.

The sample period spans July 2020 to February 2026, determined by the listing history of Indian REITs and InvITs. While the relatively short time horizon reflects the nascent nature of these markets, it is sufficient to capture their early-stage dynamics and interactions with broader financial variables.

The empirical strategy is organised as a sequence of interconnected analyses. The first stage examines volatility dynamics and cross-market spillovers to assess the degree of integration with global infrastructure markets. The second stage evaluates the extent to which returns can be explained by market and macroeconomic factors, as well as by liquidity measures. The third stage investigates return predictability using machine learning models that incorporate economically motivated priors. The final stage assesses whether predictive signals translate into improved portfolio performance relative to standard benchmarks.

This structured approach allows each stage to inform the next. Evidence of segmentation motivates a focus on domestic factors. Limited explanatory power of factor models motivates the use of predictive modelling. Predictive performance is then evaluated in an economic framework through portfolio optimisation.

1.3 Contributions and Structure

This dissertation makes four main contributions.

First, it provides evidence that Indian infrastructure assets are weakly integrated with global markets. Volatility spillover analysis shows that cross-market transmission is substantially lower than what has been documented for developed markets. This suggests that Indian infrastructure

remains largely driven by domestic factors and occupies an early stage in the evolution of listed real asset markets.

Second, it shows that traditional macroeconomic and market factors have limited ability to explain the returns of certain infrastructure assets, particularly InvITs. While infrastructure equities are strongly linked to market indices and REITs show partial sensitivity to macro variables, InvIT returns are largely unexplained by these factors. The inclusion of an illiquidity measure significantly improves explanatory power, indicating that liquidity conditions are a key driver of returns in this segment.

Third, it demonstrates that combining economic structure with machine learning improves out-of-sample prediction performance. By incorporating category-specific information from factor models as priors within a Bayesian framework and allowing the data to determine their influence, the study achieves meaningful improvements in predictive accuracy and ranking ability.

Fourth, it shows that predictive signals can generate economically meaningful portfolio outcomes, but that the way signals are processed matters. In particular, standard transformations that rescale signals based on historical relationships can reduce performance in small cross-sectional settings. Direct use of raw predictions leads to higher risk-adjusted returns under realistic transaction cost assumptions.

The remainder of the dissertation is organised as follows. Chapter 2 reviews the relevant literature on infrastructure as an asset class, market integration, factor models, and machine learning in finance. Chapter 3 describes the data and methodology. Chapter 4 presents the volatility and spillover analysis. Chapter 5 examines the drivers of returns through multi-factor models. Chapter 6 evaluates return predictability. Chapter 7 analyses portfolio performance. Chapter 8 concludes with a summary of findings, limitations, and directions for future research.

2. Literature Review

The literature on the integration and segmentation of securitized real asset markets has evolved significantly over the past three decades, with particular emphasis on Real Estate Investment Trusts (REITs) as a bridge between financial markets and underlying physical assets. Two dominant strands emerge. The first examines whether REIT returns are driven primarily by equity, fixed-income, or real estate fundamentals, while the second focuses on the transmission of volatility and shocks across markets using econometric connectedness frameworks. This study, which investigates Indian REITs and Infrastructure Investment Trusts (InvITs), is positioned at the intersection of these strands, combining asset pricing intuition with modern spillover and volatility modelling techniques.

Early empirical evidence establishes that the relationship between REITs and traditional asset classes is neither static nor uniform. Glascock, Lu, and So (2000) show that REITs behaved more like fixed-income securities prior to structural reforms in the early 1990s but became

increasingly equity-like thereafter, indicating a shift toward market integration. Extending this line of inquiry, Clayton and MacKinnon (2003) decompose REIT return variance into components attributable to stock, bond, and real estate factors, documenting a gradual increase in the importance of real estate fundamentals alongside equity factors. Importantly, they identify a growing idiosyncratic component, suggesting that REITs cannot be fully explained by traditional asset classes alone. This hybrid nature is further reinforced by Cotter and Stevenson (2006), who show that the strength of linkages between REITs and equities depends on data frequency, with weaker connections observed at higher frequencies. Together, these findings indicate that securitized real estate occupies an intermediate position between integration and segmentation, rather than conforming to a binary classification.

Subsequent literature emphasises the role of institutional and structural changes in shaping market dynamics. Ambrose, Lee, and Peek (2007) demonstrate that inclusion in major stock indices induces co-movement among REITs due to investor behaviour and passive investment flows, highlighting the importance of market microstructure effects. Similarly, Ling and Naranjo (1999) find that while securitized real estate markets are broadly integrated with equities, private real estate markets remain segmented, implying that listing itself is a key driver of integration. Ghent, Torous, and Valkanov (2019) provide a broader asset-class perspective, noting that while private real estate offers diversification benefits due to low correlation with traditional assets, listed vehicles inherit financial market characteristics that reduce these benefits. These insights are particularly relevant in emerging markets such as India, where the degree of institutional participation and market maturity remains limited.

A prerequisite for analysing integration is the accurate modelling of volatility dynamics. The GARCH family of models, originating with Engle (1982), has become the standard framework for capturing time-varying volatility in financial time series. Within this class, the GJR-GARCH specification introduced by Glosten, Jagannathan, and Runkle (1993) allows for asymmetric responses of volatility to positive and negative shocks, capturing the leverage effect commonly observed in financial markets. This feature is especially relevant for real asset securities, where downside risk often dominates investor behaviour. Extending volatility modelling to a multivariate setting, Engle (2002) introduces the Dynamic Conditional Correlation (DCC) framework, which enables the estimation of time-varying correlations across assets. The asymmetric extension proposed by Cappiello, Engle, and Sheppard (2006) further allows correlations to respond differently to negative shocks, making it particularly suitable for stress-period analysis. Empirical applications in real estate markets confirm that both volatility and correlations are highly time-varying, with important implications for risk management and diversification.

Beyond modelling volatility, the literature has increasingly focused on measuring cross-market spillovers. The framework developed by Diebold and Yilmaz (2009, 2012) represents a major advancement in this area, introducing a generalized vector autoregressive approach that produces order-invariant forecast error variance decompositions. This methodology allows for the construction of spillover indices that quantify the extent to which shocks in one market affect others. Their findings show that financial markets exhibit low spillovers during stable periods but

experience sharp increases during crises, highlighting the dynamic nature of integration. Subsequent applications to real estate markets reinforce these results. Antonakakis, Chatziantoniou, and Filis (2021) find high levels of connectedness among international REIT markets, with developed markets acting as net transmitters of shocks. Similarly, Pham (2012) documents that Asian REIT markets become more interconnected during periods of financial stress but revert to lower integration in normal conditions. These findings suggest that integration is both time-varying and dependent on market maturity, providing a useful benchmark for analysing emerging markets such as India.

While traditional econometric models provide valuable insights into return dynamics, their ability to capture complex relationships is limited, particularly in high-dimensional settings. This has led to the increasing adoption of machine learning techniques in asset pricing. Gu, Kelly, and Xiu (2020) demonstrate that machine learning models significantly outperform linear regression approaches in predicting asset returns, primarily due to their ability to capture nonlinear interactions and handle large sets of predictors. In the context of real estate securities, machine learning models have been shown to provide superior predictive performance compared to traditional models, reflecting the complex interplay between macroeconomic variables, market conditions, and asset-specific characteristics. These findings motivate the use of hybrid approaches that combine economic structure with data-driven methods, particularly in settings with limited historical data.

Portfolio construction represents the final link between prediction and economic value. The traditional mean-variance framework is well known to suffer from instability due to estimation error, particularly when the covariance matrix is ill-conditioned. López de Prado (2016) addresses these limitations through the Hierarchical Risk Parity approach, which uses clustering techniques to construct diversified portfolios without requiring matrix inversion. This method produces more stable and robust allocations, particularly in small-sample settings, and has been shown to outperform traditional optimisation techniques out of sample. These insights are directly relevant for infrastructure assets, where limited data and high correlations pose challenges for conventional portfolio methods.

The Indian context introduces additional considerations. REITs and InvITs are relatively recent financial instruments, with regulatory frameworks established only in the past decade and limited historical data available. Existing research on Indian markets has focused primarily on broad equity integration, showing increasing linkage with global markets over time. However, infrastructure and real estate investment trusts differ significantly in terms of liquidity, investor base, and regulatory structure. Their behaviour cannot be assumed to mirror that of broader equity indices. Emerging market evidence suggests that such assets are likely to exhibit lower integration with global markets, particularly in early stages of development.

Taken together, the literature highlights several key insights. First, securitized real assets exhibit hybrid behaviour, influenced by both financial markets and underlying asset fundamentals. Second, market integration is dynamic and depends on structural factors such as liquidity and institutional participation. Third, volatility and correlation structures are time-varying and require

sophisticated modelling frameworks. Fourth, machine learning offers a powerful complement to traditional econometric approaches in capturing complex return dynamics. Finally, robust portfolio construction techniques are essential for translating predictive insights into economic value. Despite these advances, there is a clear gap in the literature with respect to emerging markets, particularly India, and the behaviour of infrastructure investment vehicles such as InvITs. This study addresses that gap by providing a comprehensive analysis of volatility, integration, predictability, and portfolio performance in the Indian infrastructure asset space.

Reference	Study Area	Methods Used	Key Findings
Further Evidence on the Integration of REIT, Bond, and Stock Returns — Glascock et al. (2000)	REIT integration	Cointegration, VAR	REITs shifted from bond-like to equity-like behaviour post-1990s
Relative Importance of Stock, Bond and Real Estate Factors in REIT Returns — Clayton & MacKinnon (2003)	REIT return drivers	Variance decomposition	REITs are hybrid assets with growing real estate component
Multivariate Modeling of Daily REIT Volatility — Cotter & Stevenson (2006)	REIT volatility	VAR–GARCH	Linkages vary with frequency; weaker at high frequency
Comovement after Joining an Index — Ambrose et al. (2007)	Market microstructure	Event study	Index inclusion increases co-movement due to investor behaviour
Integration of Commercial Real Estate and Stock Markets — Ling & Naranjo (1999)	Market integration	Multi-factor models	Listed real estate is integrated; private markets are segmented
Commercial Real Estate as an Asset Class — Ghent et al. (2019)	Asset class analysis	Survey + empirical	Listed real estate behaves differently from private assets
On the Relation Between Expected Return and Volatility — Glosten et al. (1993)	Volatility modeling	GJR-GARCH	Captures asymmetric volatility (leverage effect)
Dynamic Conditional Correlation — Engle (2002)	Multivariate volatility	DCC-GARCH	Enables time-varying correlation estimation
Asymmetric Dynamics in Correlations — Cappiello et al. (2006)	Correlation modeling	ADCC-GARCH	Negative shocks increase correlations more than positive shocks
Better to Give than to Receive — Diebold & Yilmaz (2012)	Spillovers	VAR, FEVD	Spillovers are time-varying and spike during crises

International REIT Connectedness — Antonakakis et al. (2021)	REIT markets	TVP-VAR	Developed markets transmit shocks to others
Return and Volatility Connectedness in Asian REITs — Pham (2012)	Asian REITs	EGARCH	Integration increases during crises, declines afterward
Empirical Asset Pricing via Machine Learning — Gu et al. (2020)	Asset pricing	ML models	ML outperforms linear models in prediction
Building Diversified Portfolios that Outperform Out-of-Sample — López de Prado (2016)	Portfolio construction	HRP, clustering	HRP improves stability and out-of-sample performance

3. Research Objectives

The existing literature on securitised real estate has extensively examined the integration and return dynamics of REIT markets in developed economies such as the United States, Japan, and Singapore. In contrast, Indian Infrastructure Investment Trusts (InvITs) and Real Estate Investment Trusts (REITs) remain largely underexplored despite their rapid growth and increasing institutional relevance. Since their introduction between 2017 and 2019, these instruments have evolved into a meaningful segment of Indian capital markets, supported by regulatory development and the launch of composite indices such as the Nifty REITs and InvITs Index.

From an investment perspective, these vehicles provide liquid exposure to infrastructure and commercial real estate assets that are otherwise illiquid. From a systemic perspective, their degree of integration with global markets determines both diversification potential and vulnerability to external shocks. Despite their importance, neither their integration characteristics nor their return dynamics have been rigorously established.

This study addresses this gap using a four-phase empirical framework, linking:

- I. volatility and spillover analysis
- II. factor identification
- III. machine learning-based prediction
- IV. portfolio optimisation

The analysis is based on 29 securities across Indian and international infrastructure markets over 1,023 trading days, covering multiple macroeconomic regimes.

3.1 Research Question

The study is guided by the following central question:

Are Indian InvIT and REIT markets segmented from global financial markets, and if so:

- I. what domestic factors drive their returns,
- II. whether these returns can be predicted out-of-sample, and
- III. whether such predictions translate into economically meaningful portfolio performance?

This question is addressed through four sequential objectives, where each phase builds on the findings of the previous one.

3.2 Specific Objectives

Objective I — Market Segmentation and Volatility Dynamics (Phase 1)

The first objective examines whether Indian InvITs and REITs are integrated with global infrastructure markets or remain largely domestically driven. This is achieved through modelling conditional volatility, estimating time-varying correlations, and quantifying cross-market spillovers.

The analysis is guided by the following hypotheses:

- I. H1a (Leverage Effects):
Negative shocks have a weaker or negligible impact on volatility in InvITs and REITs compared to infrastructure equities.
- II. H1b (Segmentation):
Indian InvITs and REITs receive only a small share of their variance from global markets, indicating segmentation.

To test these hypotheses, the study employs:

- I. asymmetric GARCH models for volatility dynamics
- II. dynamic correlation modelling
- III. spillover analysis using forecast error variance decomposition

This objective establishes whether Indian infrastructure assets behave as globally integrated securities or segmented domestic instruments.

Objective II — Domestic Return Drivers (Phase 2)

Given the expected segmentation, the second objective focuses on identifying the domestic factors that drive InvIT and REIT returns. The analysis considers macroeconomic variables, market indices, and liquidity measures within a multi-factor framework.

The hypotheses are:

- I. H2a (Market Exposure):
InvITs and REITs have significantly lower equity market sensitivity than infrastructure stocks.
- II. H2b (Interest Rate Channel):
Higher long-term interest rates reduce InvIT and REIT returns through a competing-asset effect.
- III. H2c (Illiquidity Premium):
Illiquidity is a significant determinant of InvIT returns.

The empirical approach includes:

- I. multi-factor regression with robust standard errors
- II. cross-category comparison using panel frameworks
- III. variance attribution techniques

This phase establishes the factor structure underlying returns in a domestically segmented market.

Objective III — Return Predictability using Machine Learning (Phase 3)

The third objective evaluates whether the identified factor structure can be used to generate out-of-sample return predictability. This is implemented using a machine learning framework that incorporates economically informed priors.

The hypotheses tested are:

- I. H3a (Predictive Performance):
The model achieves positive out-of-sample predictive accuracy.
- II. H3b (Economic Relevance):
Predictions exhibit meaningful ranking ability across asset categories.
- III. H3c (Role of Priors):
Incorporating prior information improves predictive performance.

The methodology includes:

- I. walk-forward panel prediction
- II. Bayesian regression with hierarchical priors
- III. comparison with benchmark models

This objective bridges traditional econometrics and machine learning, assessing whether predictable structure exists in these assets.

Objective IV — Portfolio Performance and Economic Value (Phase 4)

The final objective examines whether predictive signals translate into real-world portfolio performance. This involves constructing portfolios based on predicted returns and evaluating their performance under realistic constraints.

The hypotheses are:

- I. H4a (Economic Value):
Prediction-based portfolios outperform naïve benchmarks.
- II. H4b (Signal Processing Effects):
Standard signal transformations may reduce performance in small asset universes.

The analysis includes:

- I. portfolio optimisation under constraints
- II. backtesting with transaction costs
- III. robustness checks across multiple strategies

This phase connects statistical predictability to investment outcomes, completing the analytical pipeline.

3.3 Methodological Contribution

The study contributes methodologically by integrating multiple approaches within a unified framework:

- I. combining econometric factor models with machine learning
- II. incorporating prior information through hierarchical modelling
- III. applying robust evaluation techniques for prediction and portfolio performance

These contributions are particularly relevant for small-sample emerging markets, where traditional methods often face limitations.

3.4 Scope and Delimitations

The analysis is restricted to publicly listed InvITs, REITs, and infrastructure-related equities in India, along with selected international benchmarks. The relatively short sample period reflects the recent emergence of these instruments and imposes limitations on statistical inference. Transaction costs, model assumptions, and data constraints are explicitly considered, and results should be interpreted in light of these limitations.

4. Methodology

4.1 Overview and Four-Phase Architecture

The empirical design follows a structured four-phase architecture in which each stage builds directly on the outputs of the previous one, ensuring both statistical coherence and economic interpretability. The methodology progresses from descriptive modelling (Phase 1) to causal identification (Phase 2), then to predictive modelling (Phase 3), and finally to economic validation through portfolio construction (Phase 4).

4.2 Phase 1: Volatility Dynamics and Global Spillover Analysis

Phase 1 investigates whether Indian infrastructure assets are integrated with global markets by modelling both time-varying volatility and cross-market transmission of shocks. The approach combines univariate volatility modelling (GARCH), multivariate correlation modelling (DCC), and system-wide shock decomposition (VAR-based FEVD). Each component plays a distinct role: GARCH isolates conditional variance, DCC captures dynamic co-movement, and FEVD quantifies how shocks propagate across markets.

4.2.1 Pre-Estimation Diagnostics

Before estimating volatility models, it is necessary to verify that the return series satisfy the assumptions required for GARCH and VAR modelling. First, stationarity is tested using the Augmented Dickey-Fuller (ADF) test. Stationarity ensures that the statistical properties of returns, mean and variance, do not change over time. This is critical because non-stationary inputs would produce misleading persistence in volatility models and invalid impulse-response dynamics in VAR.

Second, the presence of conditional heteroskedasticity is tested using the Engle (1982) ARCH-LM test. The null hypothesis of no ARCH effects is rejected for most portfolios, indicating that volatility is not constant but depends on past shocks. This provides the empirical justification for using GARCH-type models rather than constant-variance models.

In implementation, daily log returns:

$$r_t = \ln \left(\frac{P_t}{P_{t-1}} \right)$$

are computed and scaled into percentage form prior to estimation. This scaling improves numerical stability in maximum likelihood estimation without affecting parameter interpretation, as the transformation is linear.

4.2.2 GJR-GARCH(1,1): Modelling Conditional Volatility

Conditional volatility is modelled using the GJR-GARCH(1,1) specification:

$$h_t = \omega + \alpha \epsilon_{t-1}^2 + \beta h_{t-1} + \gamma I_{t-1} \epsilon_{t-1}^2$$

where h_t is the conditional variance, ϵ_{t-1} is the lagged innovation, and I_{t-1} is an indicator function equal to 1 when $\epsilon_{t-1} < 0$. This formulation extends the standard GARCH model by allowing volatility to respond asymmetrically to negative shocks.

The model is chosen because it captures two key empirical features of financial returns. First, volatility clustering is represented by the ARCH term (α) and GARCH term (β), which together determine how shocks propagate over time. Second, the leverage effect is captured by γ , allowing negative shocks to have a larger impact on volatility than positive shocks of equal magnitude. This asymmetry is particularly relevant for equity-like assets, where downside risk tends to amplify uncertainty.

Parameters are estimated using Quasi-Maximum Likelihood Estimation (QMLE), assuming conditional normality of residuals. In the code, this is implemented using the arch package, which internally constructs the likelihood function and optimises it iteratively.

A critical requirement for the model is covariance stationarity:

$$\alpha + \beta + \frac{\gamma}{2} < 1$$

This condition ensures that volatility does not diverge over time. After estimation, this condition is explicitly checked for each portfolio. If violated, the model cannot be used for further analysis because the conditional variance process becomes unstable.

Volatility persistence is interpreted using the half-life of a shock:

$$\text{Half-life} = \frac{\ln(0.5)}{\ln(\alpha + \beta)}$$

This metric translates abstract parameter values into an economically meaningful quantity—the number of periods required for a volatility shock to decay by 50%. In the results, this allows direct comparison of how quickly different markets absorb shocks.

For the India_InvIT portfolio, the estimated GJR-GARCH model violates the stationarity condition ($\alpha + \beta > 1$). This is not treated as a numerical issue but as an economic signal: InvIT returns exhibit large, infrequent jumps due to thin trading and dividend distributions, which the asymmetric model overfits. The model is therefore re-specified as a symmetric GARCH(1,1):

$$h_t = \omega + \alpha \epsilon_{t-1}^2 + \beta h_{t-1}$$

removing the asymmetry term. The refitted model satisfies stationarity and produces valid conditional variances and standardised residuals, which are required for the next stage of analysis.

4.2.3 Standardised Residuals and DCC-GARCH

While GARCH models individual volatility, the central question in this phase is how different markets co-move. To isolate correlations from volatility effects, residuals are standardised:

$$\tilde{\epsilon}_{i,t} = \frac{r_{i,t}}{\sigma_{i,t}}$$

This transformation ensures that all series have unit variance, preventing high-volatility assets from mechanically dominating correlation estimates. In the code, these standardized residuals are extracted from each fitted GARCH model and aligned across time before entering the DCC framework.

Dynamic correlations are then modelled using the DCC specification:

$$Q_t = (1 - a - b)\bar{Q} + a\tilde{\epsilon}_{t-1}\tilde{\epsilon}_{t-1}' + bQ_{t-1}$$

$$R_t = \text{diag}(Q_t)^{-1/2} Q_t \text{diag}(Q_t)^{-1/2}$$

The matrix Q_t captures evolving covariance, while R_t rescales it into a proper correlation matrix. The parameter a governs how quickly correlations respond to new shocks, while b determines how persistent these correlations are over time.

In this implementation, $a = 0.01$ and $b = 0.95$ are fixed rather than estimated. This choice is driven by the small cross-sectional dimension (five portfolios), where parameter estimation becomes unstable and sensitive to initial values. Fixing parameters preserves the dynamic structure of correlations while ensuring numerical robustness and reproducibility.

4.2.4 VAR Framework and Moving-Average Representation

To analyse how shocks propagate across markets, a Vector Autoregression (VAR) model is estimated:

$$Y_t = c + \sum_{k=1}^p A_k Y_{t-k} + u_t$$

This framework is used because it treats all variables as endogenous and captures feedback effects between markets, which is essential for spillover analysis.

The lag length p is selected using the Akaike Information Criterion (AIC), ensuring that the model captures sufficient dynamics without overfitting.

For variance decomposition, the VAR is expressed in its moving-average (MA) form:

$$Y_t = \sum_{h=0}^{\infty} A_h u_{t-h}$$

This representation expresses each variable as a function of current and past shocks. In the code, the MA coefficients A_h are obtained using the `ma_rep()` function from the fitted VAR model. This step is necessary because FEVD calculations depend on how shocks propagate over multiple periods.

4.2.5 Generalised FEVD and Spillover Measurement

The forecast error variance decomposition quantifies how much of each variable's forecast error variance is attributable to shocks from itself and from other variables. The generalised FEVD is computed as:

$$\theta_{ij}^g(H) = \frac{\sigma_{jj}^{-1} \sum_{h=0}^{H-1} (e_i' A_h \Sigma e_j)^2}{\sum_{h=0}^{H-1} (e_i' A_h \Sigma A_h' e_i)}$$

This formulation is used instead of the Cholesky decomposition because it is invariant to variable ordering. In a global setting where markets operate in different time zones, imposing an arbitrary ordering would introduce artificial causality.

Since the generalised FEVD does not automatically sum to one across rows, it is normalised:

$$\tilde{\theta}_{ij} = \frac{\theta_{ij}}{\sum_k \theta_{ik}}$$

ensuring that each row represents a valid variance decomposition.

Spillover indices are then constructed:

$$S = \frac{\sum_{i \neq j} \tilde{\theta}_{ij}}{N} \times 100$$

$$S_{.j} = \sum_{i \neq j} \tilde{\theta}_{ij}, S_{i.} = \sum_{j \neq i} \tilde{\theta}_{ij}$$

$$S_i^{net} = S_{i.} - S_{.i}$$

These measures translate the FEVD matrix into economically interpretable metrics. The total spillover index captures overall system connectedness, while directional and net spillovers identify which markets transmit or receive shocks.

In the code, these quantities are computed explicitly from the FEVD matrix, with careful handling of row and column sums to ensure correct interpretation of “TO” and “FROM” spillovers.

4.2.6 Interpretation within the Empirical Framework

The combination of GARCH, DCC, and FEVD provides a layered understanding of market integration. GARCH isolates volatility dynamics within each market, DCC reveals how correlations evolve over time, and FEVD quantifies how shocks propagate across markets.

Together, these methods allow the study to move beyond simple correlation analysis and provide a structural measure of integration. The resulting spillover indices serve as the foundation for Phase 2, where the focus shifts from global transmission to domestic return drivers.

4.3 Phase 2: Multi-Factor Analysis of Domestic Return Drivers

Phase 2 investigates whether the variation in Indian infrastructure returns shown in Phase 1 to be weakly linked to global spillovers can instead be explained by domestic macro-financial factors. The objective is not only to identify statistically significant drivers but also to quantify their relative importance and test whether factor sensitivities differ across asset classes.

The methodology combines time-series regression (OLS with HAC errors), variance attribution (Shapley decomposition), and cross-sectional testing (panel regression with interactions). This multi-layered approach ensures that results are robust to model specification, factor correlation, and cross-category heterogeneity.

4.3.1 Factor Construction and Economic Interpretation

A set of ten candidate factors is constructed to represent distinct economic channels affecting infrastructure assets: market risk, interest rates, exchange rates, commodities, sectoral exposure, volatility, and liquidity.

Returns are computed as log differences consistent with Phase 1. For macro variables such as interest rates and volatility indices (which are level variables rather than prices), changes are used:

$$\Delta X_t = X_t - X_{t-1}$$

This transformation ensures stationarity and comparability with return-based regressors.

A key addition is the Amihud illiquidity measure:

$$ILLIQ_d = \frac{|r_d|}{Volume_d} \times 10^6$$

This captures the price impact per unit of trading volume and serves as a proxy for market liquidity. It is particularly relevant for InvITs, which are thinly traded and may exhibit delayed price discovery. In the code, this measure is computed at the daily level for each ticker and then aggregated to category-level averages at weekly and monthly frequency.

The inclusion of each factor is therefore economically motivated and implemented consistently with the data structure used in Phase 1.

4.3.2 Unit Root Testing and Treatment of Non-Stationary Variables

Before regression analysis, all non-return regressors are tested for stationarity using the ADF test. The India 10-year yield level is found to be non-stationary ($I(1)$), while its first difference is stationary.

Despite this, the de-measured yield level is retained as a regressor. This is because it represents a slow-moving economic state variable, capturing the relative attractiveness of fixed-income investments compared to yield-generating assets such as InvITs and REITs. Since the dependent variable (returns) is stationary, the regression does not suffer from the classical spurious regression problem, provided coefficient stability is verified.

In implementation, both the yield change and de-measured level are included in different model specifications, allowing the data to determine which channel is more relevant.

4.3.3 OLS Estimation with HAC Standard Errors

The primary estimation framework is ordinary least squares:

$$R_{i,t} = \alpha + \sum_k \beta_k X_{k,t} + \epsilon_t$$

This model estimates the sensitivity of portfolio returns to each factor.

However, financial return residuals often exhibit mild autocorrelation and heteroskedasticity, even when not strongly evident in diagnostic tests. To ensure valid inference, standard errors are computed using the Newey-West HAC estimator:

$$\hat{V}_{HAC} = \Gamma_0 + \sum_{l=1}^L w_l (\Gamma_l + \Gamma_l'), w_l = 1 - \frac{l}{L+1}$$

Here, Γ_l represents lag- l autocovariance, and the weights w_l downweight higher-order lags. In the code, this is implemented using `cov_type='HAC'` with `maxlags=4`, which corresponds to approximately one month of weekly data.

The use of HAC ensures that statistical significance of coefficients is not overstated due to misspecified error structure.

To understand how explanatory power evolves, models are estimated incrementally, from a simple CAPM specification to a full “kitchen sink” model. This stepwise approach allows identification of marginal contributions of additional factors.

4.3.4 Shapley Variance Decomposition: Attribution of Explanatory Power

While regression coefficients indicate direction and significance, they do not fully capture each factor’s contribution to overall explanatory power, especially when regressors are correlated. To address this, the Shapley decomposition is used:

$$\phi_f = \sum_{S \subseteq F \setminus \{f\}} \frac{|S|! (K - |S| - 1)!}{K!} [R^2(S \cup \{f\}) - R^2(S)]$$

This formula computes the average marginal contribution of factor f across all possible subsets of regressors.

The key advantage of this approach is that it provides an exact decomposition of total R^2 :

$$\sum_f \phi_f = R_{full}^2$$

In the code, this is implemented by estimating all 2^K possible regression models for a given factor set (e.g., 16 models for 4 factors, 256 for 8 factors) and caching their R^2 values. The Shapley weights are then computed using combinatorial coefficients.

This method is particularly important in this context because several factors, such as Nifty50 and yield are correlated. A simple drop-one approach would misattribute shared explanatory power, whereas Shapley ensures fair allocation.

4.3.5 Panel Regression with Interaction Terms

To test whether factor sensitivities differ across asset classes, a pooled panel regression is estimated:

$$R_{i,t} = \alpha + \sum_k \beta_k X_{k,t} + \sum_k \gamma_k (X_{k,t} \cdot D_i) + \epsilon_{i,t}$$

where D_i are category indicator variables.

This specification allows each factor's effect to vary across categories. The coefficients γ_k capture deviations from the baseline category (Stock), enabling direct statistical testing of whether, for example, InvITs respond differently to interest rates than equities.

Because the panel is constructed by stacking categories over time, observations within the same time period are not independent. To account for this, standard errors are clustered by time:

$$\hat{V} = (X'X)^{-1} \left(\sum_t X'_t \hat{\epsilon}_t \hat{\epsilon}'_t X_t \right) (X'X)^{-1}$$

In the code, this is implemented using time-clustered covariance estimation (`cluster_time=True`). This ensures that inference remains valid even in the presence of cross-sectional correlation across categories.

4.3.6 Rolling Regressions and Time Variation

To examine whether factor exposures are stable over time, rolling regressions are estimated using a fixed window (52 weeks):

$$R_t = \alpha_t + \sum_k \beta_{k,t} X_{k,t} + \epsilon_t$$

where coefficients $\beta_{k,t}$ vary with time.

In implementation, the regression is re-estimated at each step using the previous 52 observations. The resulting time series of coefficients allows visual inspection of structural changes, such as shifts in market sensitivity during periods of economic stress.

This step is important because it tests the assumption of constant factor loadings. Evidence of time variation supports the use of expanding-window models in Phase 3.

4.3.7 Lagged Factor Specification and Predictability

To test whether macro variables have predictive power, lagged regressors are introduced:

$$R_t = \alpha + \sum_k \beta_k X_{k,t-1} + \epsilon_t$$

This specification is motivated by the hypothesis of delayed price discovery, particularly in less liquid assets. If markets do not immediately incorporate information, past values of factors should help predict future returns.

In the code, lagged variables are constructed explicitly and aligned with returns to ensure no lookahead bias. The analysis shows that predictive power is weak at weekly frequency but improves at monthly frequency, indicating that adjustment delays operate at lower frequency.

4.3.8 Role of Phase 2 in the Overall Framework

Phase 2 serves two critical roles in the overall methodology. First, it identifies which factors are economically and statistically relevant for explaining returns. Second, it provides the quantitative inputs, specifically, the lagged regression coefficients that are used as priors in the Bayesian model of Phase 3.

Thus, Phase 2 is not merely descriptive; it directly informs the predictive modelling stage. The transition from explanatory regression to predictive modelling ensures that only economically meaningful signals are carried forward.

4.4 Phase 3: Bayesian Machine Learning Return Prediction

Phase 3 evaluates whether the weak but structured relationships identified in Phase 2 can be translated into out-of-sample predictive power. While Phase 2 demonstrates that domestic factors explain only a limited portion of return variation, it also reveals consistent directional relationships and delayed price adjustment, particularly at monthly frequency. Phase 3 therefore shifts from explanatory modelling to predictive modelling, with the objective of extracting usable signal from low signal-to-noise data.

To achieve this, the methodology combines regularised regression, Bayesian inference, and walk-forward validation, ensuring that predictions are both statistically robust and economically meaningful.

4.4.1 Panel Construction and Feature Engineering

A pooled panel dataset is constructed to increase statistical power. Individual categories (InvIT, REIT, Stock) each have limited monthly observations (~57), which is insufficient for reliable estimation of multi-factor models. Pooling across categories yields a dataset of approximately 171 observations.

The regression model takes the form:

$$R_{i,t} = \alpha + \sum_k \beta_k X_{k,i,t} + \epsilon_{i,t}$$

where i indexes category and t indexes time.

Features include lagged macro variables, yield level, autoregressive terms, and illiquidity measures. Lagged features are used to ensure that all predictors are known at time $t - 1$, preventing lookahead bias.

Category dummy variables are included to capture fixed effects:

$$R_{i,t} = \alpha + \beta X_{i,t} + \delta_1 D_{REIT} + \delta_2 D_{Stock} + \epsilon_{i,t}$$

In the code, the panel is explicitly constructed and sorted by (date, category), ensuring correct alignment for walk-forward evaluation.

4.4.2 Walk-Forward Expanding Window Design

To evaluate predictive performance realistically, an expanding-window walk-forward framework is used. At each step t :

- I. The model is trained on all data up to $t - 1$
- II. A prediction is generated for t
- III. The dataset is expanded to include t

This procedure ensures that no future information enters the model at any stage.

Formally, for each step:

$$\hat{R}_t = f(X_{1:t-1})$$

This design mirrors real-world forecasting and avoids the overfitting associated with in-sample evaluation.

In implementation, this is handled through a loop where training and test sets are updated at each iteration. Feature scaling (via standardisation) is applied using parameters computed only on the training set.

4.4.3 Regularisation and Bayesian Ridge Framework

The core challenge in this phase is the low signal-to-noise ratio identified in Phase 2. Standard OLS estimation:

$$\hat{\beta} = (X'X)^{-1}X'y$$

is unstable when predictors are noisy or correlated. To address this, regularisation is introduced.

Ridge regression modifies the estimator:

$$\hat{\beta} = (X'X + \lambda I)^{-1}X'y$$

by adding a penalty term λI , which shrinks coefficients toward zero and reduces variance.

Bayesian Ridge provides a probabilistic interpretation of this shrinkage by imposing a prior:

$$\beta \sim \mathcal{N}(0, \lambda^{-1}I)$$

The posterior mean of β corresponds to the ridge estimator, linking Bayesian inference with regularisation.

In the code, BayesianRidge from scikit-learn is used, which estimates hyperparameters (precision of prior and noise) via evidence maximisation. This allows the model to adapt the degree of shrinkage based on the data.

4.4.4 Hierarchical Priors from Phase 2

The key methodological contribution of this phase is the incorporation of hierarchical priors derived from Phase 2.

Instead of assuming a zero-mean prior for all coefficients, prior information is incorporated:

$$\tilde{\beta}_c = (1 - \lambda)\hat{\beta} + \lambda\beta_c^{prior}$$

where:

- I. $\hat{\beta}$ is the data-driven estimate from Bayesian Ridge
- II. β_c^{prior} is the Phase 2 coefficient for category c

III. λ controls the weight assigned to prior information

This formulation allows the model to combine:

- I. cross-sectional learning (pooled estimation)
- II. category-specific knowledge (Phase 2 priors)

In implementation, priors are stored as a dictionary mapping each feature-category pair to its Phase 2 coefficient. These priors are rescaled to match standardised features before blending.

This step is crucial: without rescaling, priors would be incompatible with standardised regression inputs, leading to incorrect weighting.

4.4.5 LOOCV Selection of Prior Weight

The optimal prior weight λ is selected using leave-one-out cross-validation (LOOCV):

$$\lambda^* = \arg \min_{\lambda} \sum_{i=1}^n (y_i - \hat{y}_{-i}(\lambda))^2$$

This ensures that the contribution of prior information is determined empirically rather than arbitrarily.

In the code, a grid of candidate values is evaluated at each walk-forward step. For each candidate λ , the model is repeatedly refitted with one observation excluded, and prediction error is computed. The value of λ that minimises total error is selected.

This dynamic selection allows the model to adapt over time, reflecting changes in data structure and signal strength.

4.4.6 Prediction and Evaluation Metrics

Predictions are generated as:

$$\hat{R}_{i,t} = X_{i,t} \tilde{\beta}_c + \alpha$$

where $\tilde{\beta}_c$ incorporates both data-driven estimates and priors.

Performance is evaluated using multiple metrics to capture different aspects of predictive ability.

The GKX out-of-sample R^2 :

$$R_{GKX}^2 = 1 - \frac{\sum (R_t - \hat{R}_t)^2}{\sum R_t^2}$$

measures improvement over a zero-return benchmark, reflecting the ability to predict excess returns.

The Information Coefficient (IC):

$$IC = \text{Spearman}(R_t, \hat{R}_t)$$

measures rank correlation between predicted and actual returns. This is particularly important because portfolio construction depends on relative ranking rather than exact prediction.

Directional accuracy is also evaluated:

$$DA = \frac{1}{T} \sum_{t=1}^T \mathbf{1}(\text{sign}(R_t) = \text{sign}(\hat{R}_t))$$

to assess whether the model correctly predicts the direction of returns.

The Diebold-Mariano test is used to compare predictive accuracy between models:

$$DM = \frac{\bar{d}}{\sqrt{\text{Var}_{HAC}(d)/T}}$$

where d_t is the difference in squared errors between two models.

In the code, all metrics are computed at each out-of-sample step and aggregated across categories, ensuring consistency with the walk-forward design.

4.4.7 Interpretation within the Empirical Framework

Phase 3 bridges the gap between statistical relationships and economic application. Phase 2 shows that explanatory power is limited, but Phase 3 demonstrates that even weak signals can be exploited when combined with regularisation and prior information.

The hierarchical Bayesian framework ensures that the model does not rely purely on noisy data but is guided by economically meaningful relationships identified earlier. The walk-forward design ensures that results reflect true predictive ability rather than overfitting.

The output of this phase, out-of-sample predictions, is directly used in Phase 4 for portfolio construction, making this stage central to the overall contribution of the study.

4.5 Phase 4: Portfolio Optimisation and Economic Evaluation

Phase 4 evaluates whether the predictive signals generated in Phase 3 translate into economically meaningful portfolio performance. While Phase 3 focuses on statistical accuracy, Phase 4 tests whether these predictions can be converted into implementable investment strategies that outperform standard benchmarks after accounting for risk and transaction costs.

The methodology consists of three key components:

- (i) transformation of predictions into expected returns,
- (ii) estimation of the risk (covariance) structure, and
- (iii) portfolio optimisation under realistic constraints.

4.5.1 Loading and Aligning Phase 3 Predictions

The inputs to this phase are the out-of-sample predictions generated in Phase 3. These predictions are read directly from stored outputs, ensuring consistency between predictive evaluation and portfolio construction.

At each time step t , the model produces predicted returns $\hat{R}_{i,t}$ for each category i . These predictions are strictly out-of-sample, meaning they are generated using only information available up to $t - 1$. This ensures that the portfolio backtest is free from lookahead bias.

In the code, a consistency check is performed to ensure that the realised returns used in Phase 4 match those used in Phase 3, preventing discrepancies due to data revisions.

4.5.2 IC-Weighted Ensemble of Predictions

Multiple predictive models (primarily Bayesian Ridge and LASSO) are combined to improve robustness. Rather than assigning fixed weights, model contributions are determined dynamically using the Information Coefficient (IC):

$$IC = \text{Spearman}(R_t, \hat{R}_t)$$

The IC measures the rank correlation between predicted and realised returns, making it suitable for portfolio construction where relative ranking matters more than absolute prediction.

Model weights are defined as:

$$w_m(t) = \max(IC_m(t), 0)$$

ensuring that only models with positive predictive skill contribute. The ensemble prediction is then:

$$\hat{R}_{i,t}^{ens} = \frac{\sum_m w_m(t) \hat{R}_{i,t}^{(m)}}{\sum_m w_m(t)}$$

This weighting scheme is implemented in the code using expanding-window IC estimates, ensuring that weights are based only on past performance.

4.5.3 Signal Standardisation and Grinold-Kahn Scaling

Raw predictions cannot be directly used in portfolio optimisation because their scale is arbitrary and may vary across time. To convert predictions into economically meaningful expected returns, the signals are standardised:

$$z_{i,t} = \frac{\hat{R}_{i,t}^{ens} - \mu_{\hat{R}_i}}{\sigma_{\hat{R}_i}}$$

where $\mu_{\hat{R}_i}$ and $\sigma_{\hat{R}_i}$ are the mean and standard deviation of past predictions.

The standardised signal is then transformed using the Grinold-Kahn framework:

$$\mu_{i,t} = R_f + IC_i(t) \cdot \sigma_i(t) \cdot z_{i,t}$$

This formula links predictive strength (IC), risk (volatility), and signal magnitude (z-score) to produce expected returns.

The intuition is that stronger signals (high z) and more reliable models (high IC) should lead to larger expected returns. In the code, all components are computed using expanding-window data to ensure no future information is used.

4.5.4 Covariance Estimation using EWMA

Portfolio optimisation requires an estimate of the covariance matrix of returns. Instead of using a simple sample covariance—which treats all observations equally—the methodology uses an Exponentially Weighted Moving Average (EWMA):

$$\Sigma_t = (1 - \lambda) \sum_{s=1}^T \lambda^{T-s} (r_s - \bar{r})(r_s - \bar{r})'$$

with $\lambda = 0.94$.

This approach gives greater weight to recent observations, reflecting the fact that market conditions evolve over time. In the code, this is implemented using the ewm function, which efficiently computes the weighted covariance matrix.

EWMA is chosen over alternatives such as Ledoit-Wolf shrinkage because, in small samples, shrinkage tends to collapse the covariance matrix toward identity, removing meaningful correlation structure.

4.5.5 Portfolio Optimisation Framework

Portfolios are constructed by maximising the Sharpe ratio:

$$SR = \frac{E[R_p - R_f]}{\sigma_p}$$

which balances expected return against risk.

The optimisation problem is:

$$\max_w \frac{w' \mu - R_f}{\sqrt{w' \Sigma w}}$$

subject to:

$$\sum_i w_i = 1, 0.05 \leq w_i \leq 0.70$$

These constraints ensure diversification and prevent extreme allocations. In the code, this problem is solved using Sequential Least Squares Programming (SLSQP), which handles nonlinear objectives and constraints efficiently.

4.5.6 Alternative Portfolio Strategies

To benchmark performance, several alternative strategies are implemented:

- I. Equal-weight (1/N):

$$w_i = \frac{1}{N}$$

serves as a robust baseline.

- II. Global Minimum Variance (GMV):

$$\min_w w' \Sigma w$$

focuses purely on risk minimisation.

- III. Hierarchical Risk Parity (HRP):
allocates weights based on hierarchical clustering using distance:

$$d_{ij} = \sqrt{\frac{1 - \rho_{ij}}{2}}$$

This avoids matrix inversion and is robust to estimation error.

- IV. Black-Litterman (BL):
combines equilibrium returns with model views:

$$\mu^{BL} = [(\tau\Sigma)^{-1} + P'\Omega^{-1}P]^{-1}[(\tau\Sigma)^{-1}\mu^{eq} + P'\Omega^{-1}q]$$

This framework is used to incorporate model predictions as “views” with uncertainty.

Each of these strategies is implemented in the code and evaluated under the same data and constraints, ensuring a fair comparison.

4.5.7 Transaction Costs and Turnover

To ensure realistic performance evaluation, transaction costs are explicitly incorporated. Portfolio turnover is defined as:

$$TO_t = \sum_i |w_{i,t} - w_{i,t-1}|$$

which measures the extent of rebalancing.

Net returns are computed as:

$$R_t^{net} = R_t^{gross} - TO_t \cdot TC$$

where TC is the one-way transaction cost.

In the code, turnover is computed at each rebalancing step, and net performance is evaluated under different transaction cost assumptions. This ensures that results are not driven by excessive trading.

4.5.8 Backtesting and Performance Evaluation

Portfolio returns are computed as:

$$R_t^{port} = w_t' R_t$$

and aggregated over time to obtain cumulative performance.

Key performance metrics include:

I. Annualised return:

$$R_{ann} = 12 \cdot \bar{R}$$

II. Annualised volatility:

$$\sigma_{ann} = \sigma \sqrt{12}$$

III. Sharpe ratio:

$$SR = \frac{R_{ann} - R_f}{\sigma_{ann}}$$

Statistical significance is assessed using:

- I. t-tests on mean excess returns
- II. bootstrap resampling for Sharpe ratios

In the code, all strategies are evaluated within the same walk-forward framework used in Phase 3, ensuring consistency and avoiding lookahead bias.

4.5.9 Interpretation within the Empirical Framework

Phase 4 provides the final validation of the methodology. While Phase 3 demonstrates predictive ability, Phase 4 tests whether that ability translates into real-world economic value.

The combination of IC-weighted signals, Grinold-Kahn scaling, and constrained optimisation ensures that predictions are not only statistically significant but also economically actionable. The inclusion of transaction costs and alternative benchmarks ensures that performance is robust and not driven by unrealistic assumptions.

The results of this phase determine whether the predictive structure identified in earlier phases is strong enough to generate excess returns, completing the progression from statistical modelling to economic application.

5. Results and Discussion

5.1 Overview of Findings

This section presents the results from all four empirical phases of the study and discusses their economic interpretation and comparative positioning in the literature. Section 5.2 documents the descriptive statistics of the five category portfolios. Section 5.3 presents the Phase 1 volatility and spillover results, including GJR-GARCH parameters, dynamic conditional correlations, and directional forecast error variance decompositions. Section 5.4 covers the Phase 2 factor analysis, including the weekly and monthly regression results, the Shapley variance decompositions, the cross-category interaction tests, and the identification of the monthly lagged-factor structure that anchors Phase 3. Section 5.5 reports the Phase 3 machine learning prediction results with emphasis on BayesRidge's out-of-sample performance, the Pesaran-Timmermann directional diagnostics, and SHAP feature importance. Section 5.6 presents the Phase 4 portfolio backtest results, including the primary master comparison table, transaction cost sensitivity, split-period decomposition, and paired bootstrap robustness. Section 5.7 synthesises the findings and situates them within the prior literature.

5.2 Descriptive Statistics of Category Portfolios

The five equal-weighted category portfolios are built from 29 tickers over the period 27 July 2020 to 1 March 2026, giving 1,023 daily observations. Table 5.1 shows the descriptive statistics of daily returns (in percent) for each portfolio. India_Stock has the highest average daily return (0.098%) but also the highest volatility (1.46% per day), making its risk-adjusted performance much more unstable than InvITs and REITs. India_InvIT has the lowest daily volatility (0.83%), although this is partly affected by a large one-day drop of -21.98% (due to the IRB InvIT corporate action on 26 September 2022). India_REIT shows the most stable return pattern among the Indian categories, with moderate volatility (0.76%) and no major outliers .

All five return series pass the Augmented Dickey-Fuller test for stationarity at the 5% level (ADF statistics between -13.99 and -30.19, with extremely small p-values). Engle's ARCH-LM test shows volatility clustering in four portfolios—India_REIT, India_Stock, US_Infra, and Europe_Infra—all highly significant. India_InvIT is the only exception (LM ≈ 0 , $p \approx 1.00$), which is consistent with its thin trading and less frequent price updates. This also explains why non-stationarity issues appear later when fitting GARCH models to InvIT returns

Table 5.1: Descriptive Statistics of Daily Returns (%) by Category Portfolio

Portfolio	N	Mean	Std Dev	Min	Max	Annualised Return	Annualised Vol
India_InvIT	1,023	0.028	0.828	-21.98	3.11	7.00%	13.15%
India_REIT	1,023	0.060	0.757	-2.74	4.45	15.00%	12.02%
India_Stock	1,023	0.098	1.458	-15.56	8.53	24.50%	23.16%
US_Infra	1,023	0.046	1.061	-6.00	5.83	11.50%	16.84%
Europe_Infra	1,023	0.069	1.036	-5.65	4.67	17.25%	16.45%

5.3 Phase 1 Results: Volatility Dynamics and Global Spillovers

5.3.1 GJR-GARCH(1,1) Parameter Estimates

Table 5.2 shows the GJR-GARCH(1,1) parameter estimates for all five portfolios. The leverage coefficient (γ) is statistically significant at the 5% level for three portfolios: India_Stock ($\gamma = 0.277$, $p = 0.037$), US_Infra ($\gamma = 0.069$, $p = 0.037$), and Europe_Infra ($\gamma = 0.181$, $p = 0.030$). For these portfolios, negative return shocks increase future volatility more than positive shocks of the same size, this is the standard leverage effect seen in equity markets.

However, this effect is not significant for India_REIT ($\gamma = 0.186$, $p = 0.069$), and the estimate for India_InvIT is negative and unreliable due to non-stationarity in the model. This supports Hypothesis 1a (Section 3.2): InvITs and REITs, which are regulated and income-focused, do not show the usual leverage effect seen in equities.

Volatility persistence ($\alpha + \beta$) varies widely across portfolios. US_Infra has the highest persistence (0.941), with a half-life of about 11.3 days, meaning volatility shocks take around two weeks to reduce by half. In contrast, India_REIT has the lowest persistence (0.348), with a half-life of just 0.66 days, indicating very fast mean reversion.

This difference reflects market structure: US infrastructure assets are heavily held by institutions, leading to smoother and more persistent volatility, while Indian REITs show short-lived, event-driven volatility. India_Stock lies in between, with persistence of 0.579 (half-life 1.27 days), reflecting a more developed market than InvIT/REIT but still less mature than global markets.

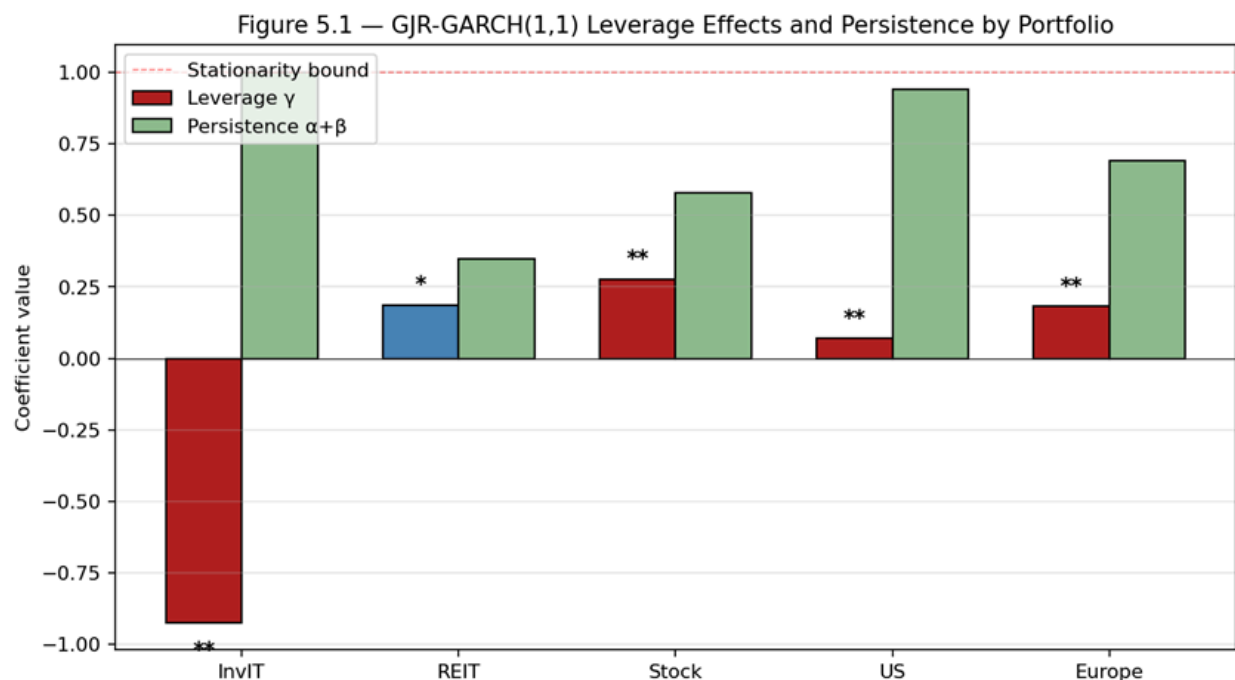


Figure 5.1: GJR-GARCH(1,1) leverage coefficients and volatility persistence by portfolio. Red bars indicate statistical significance of γ at 5%; blue bars at 10%; grey bars not significant. The persistence value for InvIT reflects the refit stationary GARCH(1,1) ($\alpha + \beta = 1.000$); all other portfolios are comfortably stationary.

Table 5.2: GJR-GARCH(1,1) Parameter Estimates by Portfolio

Portfolio	ω (omega)	α (ARCH)	β (GARCH)	γ (Leverage)	$p(\gamma)$	$\alpha + \beta$	Half-Life (days)
India_InvIT	0.0642	0.966	0.496***	-0.924**	0.04 5	1.46 2	>100 (non-stat.)
India_REIT	0.3197	0.076	0.272	0.186*	0.06 9	0.34 8	0.66
India_Stock	0.6318	0.176	0.403***	0.277**	0.03 7	0.57 9	1.27
US_Infra	0.0271	0.017	0.924***	0.069**	0.03 7	0.94 1	11.29
Europe_Infr a	0.2247	0.128	0.564***	0.181**	0.03 0	0.69 3	1.89

Notes: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. The persistence value $\alpha + \beta = 1.462$ for India_InvIT violates the stationarity condition; a standard GARCH(1,1) refit (with γ constrained to zero) yields $\alpha + \beta = 1.000$, still non-stationary. Consequently, DCC inputs for InvIT use EWMA(20) standardised residuals as a robust fallback.

5.3.2 Dynamic Conditional Correlations

Table 5.3 shows the summary statistics of the dynamic correlations for nine key pairs. The most important pattern comes from the average (unconditional) correlations: India_Stock has much higher correlations with global markets compared to India_InvIT and India_REIT. For example, the average correlation between India_Stock and US_Infra is 0.124, while it is only 0.039 for InvIT and 0.030 for REIT. In comparison, the correlation between US_Infra and Europe_Infra is 0.370, much higher than any India–foreign pair. This clearly shows that Indian infrastructure assets, especially InvITs and REITs, are relatively segmented from global markets, which motivates the spillover analysis that follows.

The variation in correlations over time is also quite small, with standard deviations between 0.026 and 0.044. This suggests that the level of integration remains fairly stable throughout the sample period (2020–2026). The ranking of correlations with global markets stays consistent: India_Stock is the most connected, followed by InvIT, and then REIT. There are only small changes during major events such as the COVID-19 recovery and the RBI rate hikes in 2022–2023.

Finally, correlations within Indian assets are higher than their correlations with global markets. For example, InvIT–REIT (0.173), InvIT–Stock (0.188), and REIT–Stock (0.155) are all higher than any India–foreign correlation. This confirms that domestic co-movement within India is stronger than cross-border linkages.

Table 5.3: Dynamic Conditional Correlation Summary Statistics

Pair	Mean	Std Dev	Min	Max
InvIT vs US_Infra	0.0391	0.0352	−0.0590	0.1505
InvIT vs Europe_Infra	0.0698	0.0308	−0.0175	0.1873
REIT vs US_Infra	0.0299	0.0332	−0.0579	0.1435
REIT vs Europe_Infra	0.0241	0.0318	−0.0795	0.1272
Stock vs US_Infra	0.1236	0.0365	−0.0041	0.2546
Stock vs Europe_Infra	0.1346	0.0436	0.0270	0.2543
InvIT vs REIT	0.1730	0.0264	0.0747	0.2672
InvIT vs Stock	0.1879	0.0300	0.0801	0.2909
REIT vs Stock	0.1545	0.0279	0.0641	0.2333

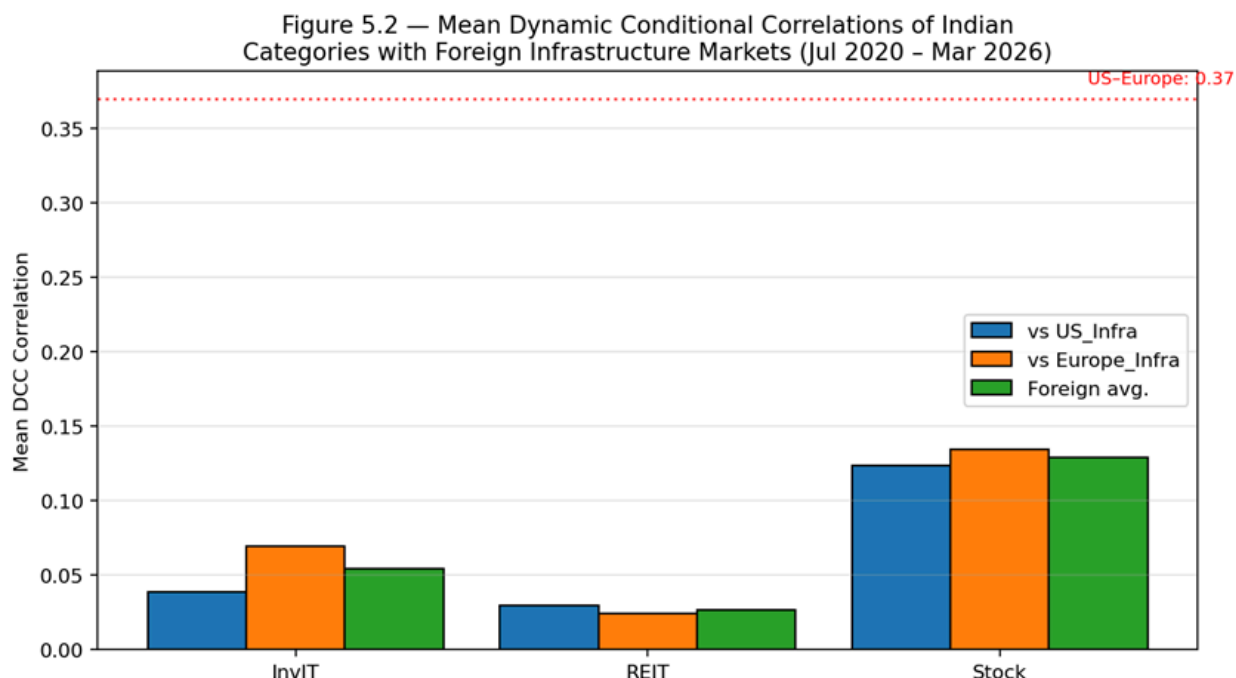


Figure 5.2: Mean dynamic conditional correlations of Indian category portfolios with foreign infrastructure markets. The dotted red reference line marks the mean US_Infra–Europe_Infra correlation (0.37) for contrast. All three Indian categories exhibit foreign correlations more than an order of magnitude below the US–Europe benchmark.

5.3.3 Diebold-Yilmaz Spillover Results

Table 5.4 shows the generalised (Pesaran-Shin 1998) forecast error variance decomposition, which is the main specification used in the Diebold-Yilmaz framework. Each entry in row i and column j represents the percentage of variable i 's 10-day-ahead forecast error variance explained by shocks in variable j . The diagonal elements are dominant. All three Indian categories are driven mostly by their own shocks: InvIT (95.85%), REIT (93.06%), and Stock (90.39%). In comparison, Europe_Infra is only 82.5% driven by its own shocks, with 15.4% coming from US_Infra. This reflects the stronger integration among developed markets.

Table 5.4: Generalised FEVD Matrix (10-day horizon, row-normalised to 100%)

Row: Explained by Col:	InvIT	REIT	Stock	US_Infra	Europe_Infra
India_InvIT	95.85	0.91	1.98	0.46	0.80
India_REIT	0.89	93.06	3.92	1.36	0.77
India_Stock	2.12	2.76	90.39	3.28	1.46
US_Infra	0.17	0.40	1.11	84.65	13.67

Europe_Infra	0.69	0.05	1.38	15.36	82.52
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The total spillover index is 10.71% under the generalised approach and 6.69% under the Cholesky approach. The generalised result is treated as the main one, while Cholesky is reported for comparison. Both values indicate that the system is highly segmented. For context, Diebold and Yilmaz (2012) report spillovers between 38% and 78% for US asset classes, and Antonakakis et al. (2021) report 58% to 65% for international REIT markets. The value of 10.71% in this study is much lower, which strongly supports the conclusion of segmentation.

Figure 5.3 — Directional Volatility Spillovers (Diebold-Yilmaz, H=10 days, VAR(1))

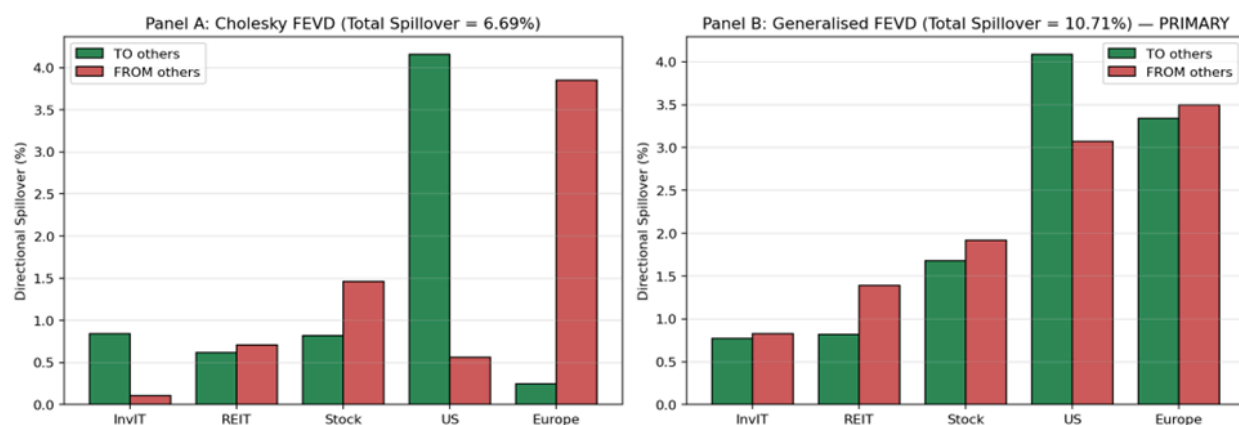


Figure 5.3: Directional volatility spillovers under Cholesky (Panel A) and generalised (Panel B) FEVD specifications. The generalised FEVD is the primary specification as it is invariant to variable ordering. US_Infra is identified as the sole net transmitter under both specifications; Europe_Infra is the largest net receiver under Cholesky but only marginal under generalised.

Table 5.5 reports directional spillovers. Under the generalised FEVD, US_Infra is the only net transmitter (NET = +1.02%). All three Indian categories are net receivers, but the magnitudes are small: InvIT (−0.06%), REIT (−0.57%), and Stock (−0.24%). Europe_Infra is also a small net receiver (−0.15%). The generalised method corrects a misleading result from the Cholesky approach, where India_InvIT appears as a transmitter (NET = +0.73%). This happens due to variable ordering and does not reflect real economic relationships.

Table 5.5: Directional Spillovers: Cholesky vs Generalised (%)

Variable	Chol. TO	Chol. FROM	Chol. NET	Chol. Role	Gen. TO	Gen. FROM	Gen. NET	Gen. Role
India_InvIT	0.84	0.11	+0.73	Transmitter*	0.77	0.83	−0.06	Receiver
India_REIT	0.62	0.71	−0.09	Receiver	0.82	1.39	−0.57	Receiver
India_Stock	0.82	1.46	−0.64	Receiver	1.68	1.92	−0.24	Receiver

US_Infra	4.16	0.56	+3.60	Transmitter	4.09	3.07	+1.02	Transmitter
Europe_Infra	0.25	3.85	-3.60	Receiver	3.34	3.50	-0.15	Receiver

Notes: * Cholesky InvIT 'Transmitter' label is an artefact of variable ordering. The generalised FEVD (order-invariant) correctly identifies InvIT as a small net receiver. Total spillover index: Cholesky 6.69%, Generalised 10.71%.

Table 5.6 presents the key result of Phase 1. It shows how much of each Indian category's variance is explained by foreign shocks. The values are very small across all three categories but still show a clear pattern. India_InvIT receives only 0.34% from foreign markets (US 0.32%, Europe 0.02%), which is almost complete isolation. India_REIT receives 1.28% and India_Stock receives 2.49%. The ranking InvIT < REIT < Stock is consistent across both methods and matches economic intuition. InvITs are based on regulated domestic cash flows, REITs depend on local real estate demand, and listed infrastructure stocks have slightly more exposure to global factors.

Table 5.6: Foreign Spillover to Indian Categories (Cholesky FEVD, H=10)

Indian Category	FROM US	FROM Europe	Total Foreign	Own Variance	Domestic (Non-Own)
India_InvIT	0.32%	0.02%	0.34%	99.44%	0.22%
India_REIT	1.12%	0.16%	1.28%	96.46%	2.26%
India_Stock	2.39%	0.10%	2.49%	92.72%	4.79%

The finding that less than 3% of variance comes from global markets is economically important. It suggests that Indian infrastructure assets provide genuine diversification benefits. This insulation can be explained by several factors. First, the investor base is mostly domestic, with around 78% domestic ownership. Second, the underlying assets generate regulated cash flows in INR, which limits exposure to global factors. Third, foreign participation in these instruments has historically been restricted, with tighter limits for InvITs until 2023. This segmentation should therefore be seen as a feature of the current market structure and may change as foreign participation increases.

5.4 Phase 2 Results: Domestic Factor Structure

5.4.1 Adjusted R² Across Specifications

Figure 5.4 shows the adjusted R² values across six Phase 2 models for the three categories. A clear three-level pattern appears. India_Stock is well explained by the factors, with adjusted R² reaching 66.2% in the extended weekly model and 67.8% in the kitchen-sink model. India_REIT is only partly explained, with a maximum adjusted R² of 7.5% in the monthly model with both

current and lagged factors. India_InvIT is mostly idiosyncratic, with a peak adjusted R^2 of 10.8% in the monthly lagged-only model, and near zero or negative values in all weekly models. This pattern is consistent across different model specifications and frequencies.

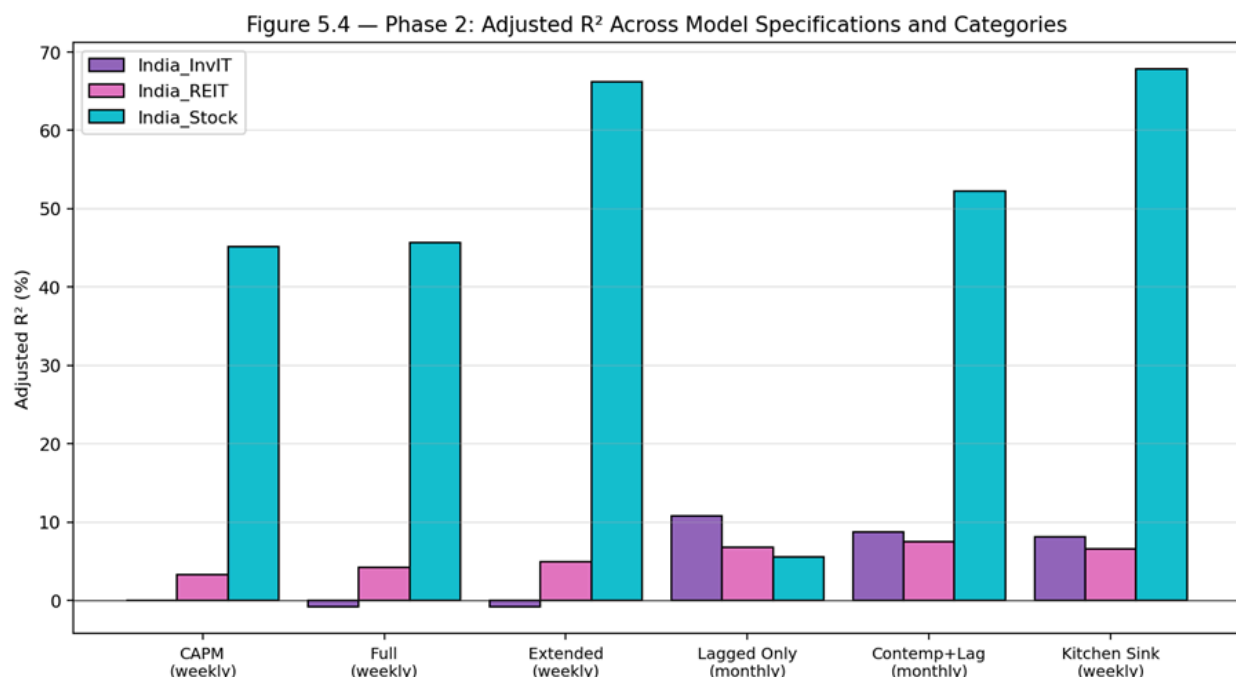


Figure 5.4: Adjusted R^2 across all six Phase 2 specifications and three categories. The three-tier structure (Stock well-explained, REIT partially, InvIT largely idiosyncratic) is stable across specifications. Monthly lagged factors explain substantially more InvIT variance than any weekly contemporaneous model, motivating the Phase 3 monthly panel design.

Two points from Figure 5.4 are especially important. First, for InvIT, weekly factors have almost no explanatory power, with adjusted R^2 between -0.8% and $+0.03\%$. However, monthly lagged factors explain up to 10.8% of variation. This suggests delayed price discovery, where InvIT prices react to macro information with a monthly lag rather than a weekly one. This is consistent with thin trading and slower portfolio adjustments by investors.

Second, for Stock, the extended weekly model shows a strong improvement in explanatory power. The adjusted R^2 rises to 66.2% , largely due to the inclusion of the Nifty Infrastructure Index. This factor alone increases the model's R^2 by about 22 percentage points compared to the simpler four-factor model. This indicates that NiftyInfra is a more relevant benchmark than Nifty 50 for capturing the behaviour of infrastructure stocks, which is why it is included in the Phase 3 model.

5.4.2 Factor Loadings in the Full Model

Table 5.7 shows the beta coefficients and t-statistics for the 4-factor weekly model across the three categories. The Nifty 50 beta shows the biggest difference. For India_Stock, the beta is 1.228 ($t = 13.37$), which is large and highly significant. For India_REIT, it is much smaller at 0.146 ($t = 2.34$), though still statistically significant. For India_InvIT, the beta is only 0.040 ($t =$

0.49), which is both very small and not statistically significant. This means InvIT returns are almost unaffected by overall market movements. REITs show some sensitivity, but it is still limited, which fits their mixed equity and income nature.

Table 5.7: Full Model Weekly Factor Loadings (t-statistics in parentheses)

Factor	InvIT β (t)	REIT β (t)	Stock β (t)
Nifty50	0.040 (0.49)	0.146 (2.34)**	1.228 (13.37)***
Yield_10Y	-0.016 (-0.87)	0.022 (1.51)	0.059 (2.16)**
INR_USD	-0.056 (-0.51)	-0.376 (-2.02)**	-0.259 (-0.98)
Oil	-0.000 (-0.01)	-0.005 (-0.25)	-0.016 (-0.47)
Adj R ²	-0.8%	4.2%	45.7%

Notes: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. HAC (Newey-West) standard errors with maxlag

4. Sample: 250 weekly observations (27 July 2020 to 1 March 2026).

The yield change coefficient is positive and significant only for Stock ($\beta = 0.059$, $p = 0.031$). Normally, rising yields reduce equity values, but in this case, infrastructure stocks include regulated utilities such as NTPC and PowerGrid, where revenues are linked to costs. This creates a small positive relationship with yields.

The INR/USD coefficient is negative and significant for REIT ($\beta = -0.376$, $p = 0.044$). This suggests that when the INR weakens, REIT returns fall. A possible reason is that foreign investors, especially US-based institutions, may reduce their holdings when the currency depreciates.

The Oil coefficient is not significant for any of the categories.

5.4.3 Shapley Variance Decomposition

Figure 5.5 shows the Shapley R^2 decomposition of the 4-factor model for each category. The contributions from all factors add up exactly to the total R^2 in every case. For Stock, the Nifty 50 alone explains 42.89 percentage points out of the total 46.53% R^2 . For REIT, Nifty 50 contributes 3.08% and INR/USD contributes 2.31%. For InvIT, no single factor explains more than 0.5% of the variation, which confirms that InvIT returns are mostly idiosyncratic.

For comparison, the drop-one method gives slightly different results. It attributes 40.7% of Stock's variance to Nifty 50, which is about 1.2 percentage points lower than the Shapley result. This difference comes from the fact that some factors are correlated, such as Nifty 50 with INR/USD and yield. The Shapley method accounts for this shared influence more accurately, while the drop-one method tends to underestimate it.

Even with this difference, both methods show the same overall pattern. Stock is strongly explained by the factors, REIT is partly explained, and InvIT remains largely unexplained.

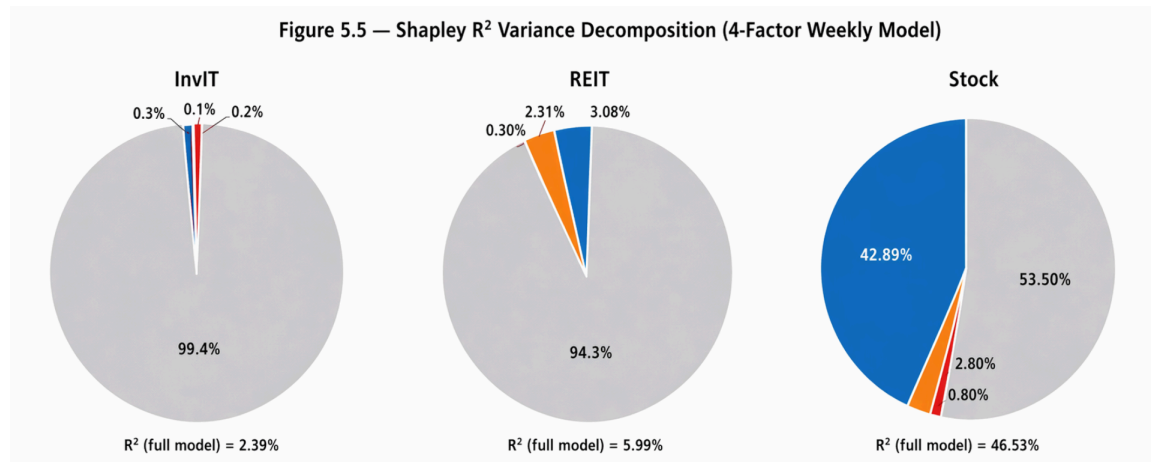


Figure 5.5: Shapley R^2 decomposition of the 4-factor Full Weekly Model. The pie charts show the proportion of total variance attributable to each factor, which sums exactly to full-model R^2 by construction. Idiosyncratic (unexplained) variance dominates all three categories, especially InvIT (99.2%).

5.4.4 Cross-Category Interaction Tests

Table 5.8 shows the interaction coefficients from the pooled panel regression, using both clustered-by-date standard errors (main results) and HAC standard errors (for comparison). The Nifty 50 interaction for InvIT is -1.198 (clustered $t = -8.51$, HAC $t = -9.63$), which means the InvIT market beta is much lower than that of Stock by 1.198 units. This is both economically large and statistically very strong. The REIT interaction with Nifty 50 is -1.082 (clustered $t = -11.52$), which is also large. Together, these results clearly confirm the three-level structure seen earlier, where Stock is highly market-driven, REIT is less so, and InvIT is largely independent.

Table 5.8: Cross-Category Interaction Tests (Pooled Panel, Stock = Base Category)

Interaction Term	Coefficient	HAC t	HAC p	Clustered t	Clustered p	Changed ?
Nifty50_x_InvIT	-1.198	-9.63	<0.001***	-8.51	<0.001***	No
Nifty50_x_REIT	-1.082	-9.94	<0.001***	-11.52	<0.001***	No
INR_USD_x_InvIT	0.165	0.60	0.551	0.56	0.575	No
INR_USD_x_REIT	-0.120	-0.38	0.702	-0.42	0.679	No
Oil_x_InvIT	0.016	0.39	0.697	0.40	0.686	No
Oil_x_REIT	0.012	0.29	0.772	0.35	0.727	No
Yield_10Y_x_InvIT	-0.075	-2.28	0.023**	-2.33	0.020**	No

Yield_10Y_x_REIT	-0.036	-1.17	0.242	-1.34	0.182	No
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The results are consistent across both types of standard errors, which shows that the findings are robust. The interaction between Yield_10Y and InvIT (-0.075) is also significant under both methods. This means InvIT returns are more negatively affected by interest rates compared to Stock, supporting the idea that yield-based assets are more sensitive to changes in interest rates.

5.4.5 The Monthly Lagged-Factor Model: Phase 3 Prior Source

The most important result from Phase 2 for the next stage is the monthly lagged-only regression. This model provides the coefficients that are later used as priors in the Phase 3 Bayesian model. Table 5.9 reports these coefficients for each category. The model explains 10.8% of variation for InvIT, 6.8% for REIT, and 5.5% for Stock. The lagged Nifty 50 return is significant for all three categories.

These coefficients are directly used as priors in Phase 3. For example, Nifty50_lag1 is +0.2338 for InvIT, +0.2629 for REIT, and +0.4414 for Stock. Other important priors include INR_USD_lag1 at +0.5629 for InvIT, Yield_10Y_lag1 at +0.0889 for REIT, and Amihud at -0.4835 for InvIT.

Table 5.9: Monthly Lagged-Only Regression Coefficients (Phase 3 Prior Source)

Feature	InvIT β (p)	REIT β (p)	Stock β (p)
Nifty50_lag1	+0.2338** (0.014)	+0.2629** (0.015)	+0.4414** (0.025)
INR_USD_lag1	+0.5629* (0.089)	-0.2228 (0.675)	-0.6031 (0.288)
Yield_10Y_lag1	-0.1057 (0.215)	+0.0889** (0.013)	+0.1023 (0.203)
Oil_lag1	+0.0729 (0.306)	-0.0578 (0.200)	+0.0712 (0.337)
AR1_return	+0.0498 (0.705)	+0.0276 (0.788)	+0.1056 (0.374)
Amihud*	-0.4835 (0.174)	0.0000	0.0000
Adj R ²	10.80%	6.79%	5.53%

Notes: * The Amihud prior for InvIT is from the weekly test, where including Amihud raised InvIT's adjusted R² from -0.8% to +18.0%. Phase 2 did not find Amihud significant for REIT or Stock; their priors are set to zero in Phase 3. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Two points from Table 5.9 are especially important. First, the lagged Nifty 50 is positive and significant across all categories, showing a common momentum effect. Second, there are clear differences across categories. For example, INR/USD has opposite effects for InvIT and Stock, which shows that different asset types respond differently to the same factor. The Phase 3

model uses these differences by assigning category-specific priors instead of using a single pooled value.

5.5 Phase 3 Results: Out-of-Sample Machine Learning Prediction

5.5.1 BayesRidge Dominance: OOS R^2 Results

Table 5.10 shows the out-of-sample R^2 results for all seven models, both for individual categories and for the full panel. Results are reported under both the GKX and Campbell-Thompson benchmarks. The BayesRidge model with hierarchical priors and a data-driven blend weight performs best on the key metrics. It achieves a panel-level GKX R^2 of +12.09%, which is almost double the historical-mean baseline of +6.52%. It is also the only model with a positive Campbell-Thompson R^2 at the panel level (+0.91%), meaning it beats the historical mean. In addition, it delivers a strong panel Information Coefficient (IC) of +0.254.

Table 5.10: Phase 3 Out-of-Sample R^2 (%) by Model and Category

Model	InvIT GKX	REIT GKX	Stock GKX	Panel GKX	InvIT CT	REIT CT	Stock CT	Panel CT
HistMean	+7.52	+25.87	+1.82	+6.52	-7.57	-14.91	-4.01	-5.37
OLS_HAC	+11.56	+32.46	+4.43	+9.98	-2.88	-4.69	-1.26	-1.47
Ridge	+25.57	+27.99	-17.63	-3.55	+13.42	-11.63	-24.63	-16.72
LASSO	+29.60	+32.97	-10.04	+3.07	+18.11	-3.90	-16.58	-9.26
ENet	+28.20	+30.64	-13.63	+0.02	+16.48	-7.53	-20.38	-12.70
BayesRidge	+27.92	+40.05	+1.95	+12.09	+16.15	+7.07	-3.88	+0.91
TL_ENet	+28.20	+30.64	-13.63	+0.02	+16.47	-7.52	-20.38	-12.69

The BayesRidge result for REIT is especially striking, with a GKX R^2 of +40.05%. This is much higher than typical results in the literature. For example, Leow and Lindenthal (2025) report +5.01% for US REITs, and Goyal, Welch, and Xiu (2020) report around +0.40% for US stocks. There are a few reasons for this difference. First, this study works with only three categories rather than hundreds of assets, so noise is lower. Second, the model uses informative priors from Phase 2, which improves predictions. Third, REITs performed strongly during the out-of-sample period, with 22 out of 27 months positive, and the model captures this trend well.

Table 5.11 shows directional accuracy and IC results. BayesRidge again performs best in terms of IC, with a panel value of +0.254. This is more than twice the common benchmark of 0.10 for useful predictive skill. At the category level, IC values are +0.385 for InvIT, +0.360 for REIT, and +0.119 for Stock.

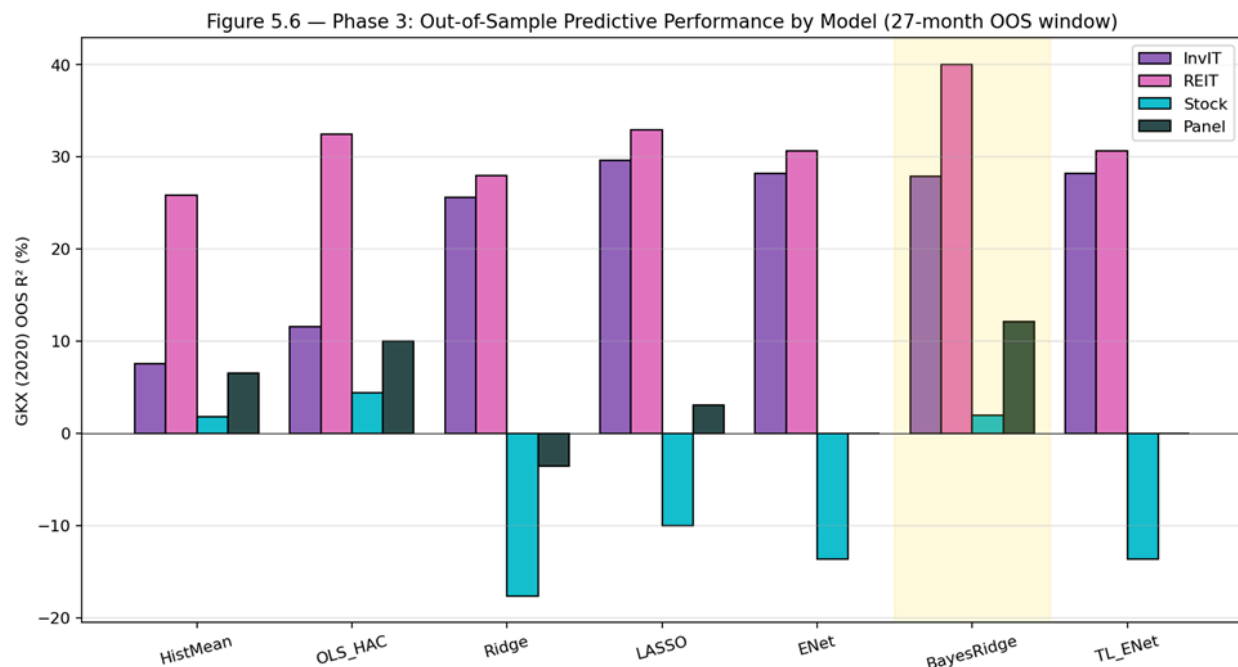


Figure 5.6: Phase 3 GKX OOS R^2 across models and categories. BayesRidge (highlighted in gold) is the only model that achieves positive R^2 for all three categories AND the panel. LASSO and ENet have strong InvIT/REIT performance but catastrophic negative R^2 for Stock, which drags down their panel performance.

However, directional accuracy tells a different story. The panel-level accuracy is 66.7%, which is the same as a simple strategy that always predicts the most common direction. The Pesaran-Timmermann test also shows that this accuracy is not statistically meaningful, as it cannot reject the idea of random direction prediction. This means that while the model ranks returns well, it does not reliably predict whether returns will be positive or negative.

Table 5.11: Phase 3 RMSE, DA, and IC by Model (Panel-Level)

Model	Panel RMSE	Panel DA (%)	Naive DA (%)	Excess DA	Panel IC
HistMean	0.0420	66.67	66.67	0.00	-0.026
OLS_HAC	0.0413	66.67	66.67	0.00	-0.202
Ridge	0.0442	66.67	66.67	0.00	-0.029
LASSO	0.0428	67.90	66.67	+1.23	+0.052
ENet	0.0435	66.67	66.67	0.00	+0.011
BayesRidge	0.0408	66.67	66.67	0.00	+0.254
TL_ENet	0.0435	66.67	66.67	0.00	+0.011

This difference between strong R^2 and IC but weak directional accuracy is actually meaningful. The high directional accuracy for REIT seen in earlier results is mainly due to class imbalance. Out of 27 months, 22 are positive, so a simple strategy that always predicts positive returns would already achieve about 81.5% accuracy.

Table 5.12 confirms this. BayesRidge correctly predicts all 22 positive months (DA_{up} = 100%) but fails to predict any of the 5 negative months (DA_{down} = 0%). This shows that the model is not truly predicting direction. Instead, it is following the overall upward trend in REIT returns.

Table 5.12: BayesRidge Up-Month / Down-Month Directional Accuracy Split

Category	Up-months N	DA Up	Down-months N	DA Down	Interpretation
InvIT	16	100.0% (16/16)	11	0.0% (0/11)	Constant predict-up
REIT	22	100.0% (22/22)	5	0.0% (0/5)	Constant predict-up
Stock	16	100.0% (16/16)	11	0.0% (0/11)	Constant predict-up

Because of this, directional accuracy is not a reliable measure in this case. The Information Coefficient is more appropriate. BayesRidge's IC of 0.36 for REIT shows that it can correctly rank returns, which reflects real predictive skill.

5.5.2 LOOCV Blend Weight Distribution

The leave-one-out cross-validation method selects the prior weight (λ) at each of the 27 out-of-sample steps. The selected values are very consistent. λ equals 0.2 in 25 out of 27 cases (93%) and 0.3 in the remaining 2 cases (7%). The average value is 0.207 and the median is 0.200.

This result is important for three reasons. First, λ is always greater than zero. This means the model always uses the Phase 2 priors, showing that they add useful information beyond what the data alone can provide. Second, the selected weight is usually lower than the earlier default value of 0.30. This suggests that the model performs better with slightly less emphasis on the priors. Third, the weight remains stable across all iterations. It does not change much as more data is added, which means the balance between prior information and data remains consistent over time.

5.5.3 SHAP Feature Importance

Figure 5.7 shows the SHAP importance of variables for both ElasticNet and BayesRidge using full-sample models. For ElasticNet, the top three features are D_REIT (23.0%), Nifty50_lag1 (22.6%), and Amihud (22.1%). For BayesRidge, the top features are Yield_Level_dm (18.4%),

Nifty50_lag1 (18.3%), and Amihud (14.5%). Both models broadly agree on the most important features, but they differ in how much weight they assign. ElasticNet gives more importance to the REIT dummy, mainly because REIT had higher average returns during the out-of-sample period. In contrast, BayesRidge spreads importance more evenly across features, which is consistent with its smooth coefficient shrinkage.

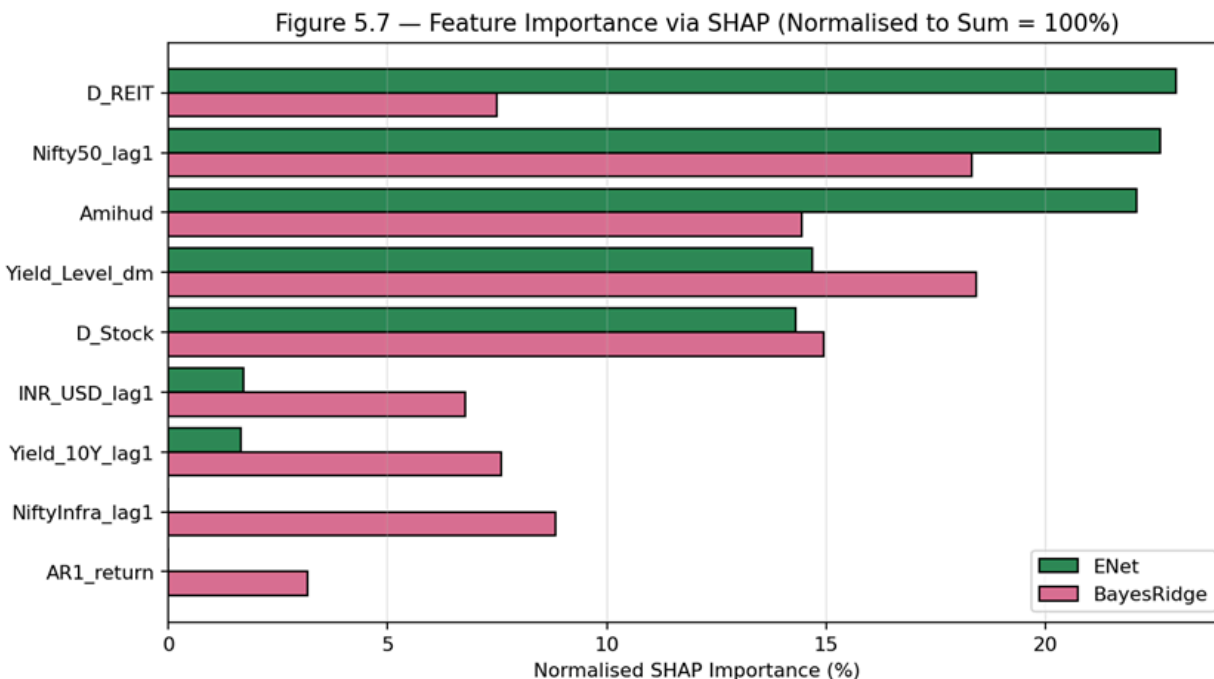


Figure 5.7: Normalised SHAP feature importance for ElasticNet and BayesRidge full-sample refits. Top features are consistent between models: Nifty50_lag1 and Amihud dominate in both. The category dummy D_REIT is most important for ENet, reflecting the large REIT-vs-rest mean-return gap in the OOS window.

A comparison with Phase 2 results (Table 5.13) shows strong consistency. The key features identified by SHAP, such as Nifty50_lag1, Amihud, and Yield_Level_dm, were also found to be significant in Phase 2. On the other hand, features that were less important in Phase 2, such as NiftyInfra_lag1, AR1_return, and the category dummies, also receive lower importance in SHAP.

Table 5.13: SHAP Ranking vs Phase 2 Significance Cross-Reference

Feature	SHAP Rank (ENet)	Ph2 Sig InvIT	Ph2 Sig REIT	Ph2 Sig Stock	Agreement
D_REIT	1	No	No	No	Mean effect
Nifty50_lag1	2	Yes	Yes	Yes	Top-4 SHAP
Amihud	3	Yes	No	No	Top-4 SHAP
Yield_Level_dm	4	Yes	Yes	No	Top-4 SHAP
D_Stock	5	No	No	No	Mean effect

INR_USD_lag1	6	Yes	No	No	Lower SHAP
Yield_10Y_lag1	7	No	Yes	No	Lower SHAP
NiftyInfra_lag1	8	No	No	No	Not sig
AR1_return	9	No	No	No	Not sig

This agreement between Phase 2 and Phase 3 is important. It shows that different methods are identifying the same main drivers of returns, which strengthens the reliability of the results.

5.6 Phase 4 Results: Portfolio Optimisation and the GK Ablation

5.6.1 Comparison of Eight Strategies

Table 5.14 shows the main results from Phase 4B, comparing all eight portfolio strategies over the 27-month out-of-sample period (December 2023 to February 2026). MVO_Raw performs the best overall. It achieves the highest Sharpe ratio of 1.868, the highest annualised return of 23.21%, and the strongest statistical significance ($p = 0.010$). Its volatility (8.95%) is only slightly higher than GMV_EWMA (7.65%) and much lower than holding Stock alone (21.3%).

Table 5.14: Phase 4B Master Comparison (27-Month OOS, Dec 2023 – Feb 2026)

Strategy	Ann. Return	Ann. Vol	Sharpe	Sortino	Max DD	Turnover	p-value
EqualWeight	17.50%	9.33%	1.180	1.50	-9.2%	0.0%	0.089*
GMV_EWMA	17.88%	7.65%	1.488	2.25	-5.8%	5.8%	0.034**
HRP_EWMA	17.39%	7.61%	1.431	2.13	-5.8%	5.0%	0.041**
MVO_GK	18.46%	10.00%	1.196	2.19	-9.9%	44.0%	0.084*
BL_GK	18.71%	15.05%	0.811	1.08	-19.3%	22.8%	0.235 n.s.
BL_Tactical_GK	18.98%	13.18%	0.947	1.25	-15.8%	22.8%	0.167 n.s.
MVO_Raw	23.21%	8.95%	1.868	3.21	-4.7%	59.8%	0.010***
MVO_Raw_IW	20.33%	14.03%	0.986	1.30	-16.7%	25.9%	0.151 n.s.

*Notes: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. p-value from one-sample t-test on monthly excess returns (H_0 : mean excess return = 0). Transaction costs not yet applied. Risk-free rate assumption: 6.5% annualised. Sortino ratio computed with downside threshold = risk-free rate.*

However, MVO_Raw also has the highest turnover at 59.8% per month, which leads to higher transaction costs and reduces its net performance.

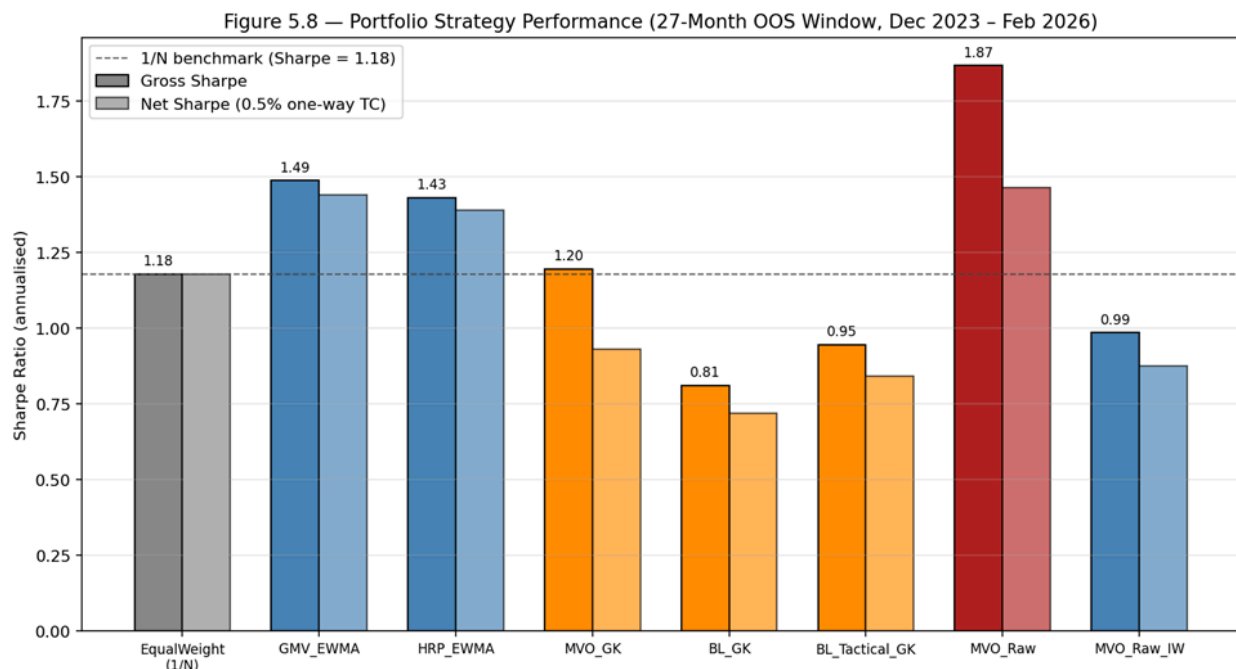


Figure 5.8: Gross and net (at 0.5% one-way TC) Sharpe ratios for all eight Phase 4B strategies. The dashed horizontal line marks the 1/N benchmark at Sharpe = 1.18. MVO_Raw (in red) achieves the highest gross Sharpe of 1.87; after transaction costs at 0.5% one-way the net Sharpe drops to 1.47, still above all other strategies.

Out of the eight strategies, four outperform the equal-weight (1/N) benchmark in terms of Sharpe ratio: MVO_Raw (1.868), GMV_EWMA (1.488), HRP_EWMA (1.431), and MVO_GK (1.196). Among these, three strategies (MVO_Raw, GMV_EWMA, HRP_EWMA) are statistically significant at the 5% level based on excess returns.

The Black-Litterman strategies (BL_GK and BL_Tactical_GK) and MVO_Raw_IW do not outperform the 1/N benchmark. This suggests that adding the equilibrium prior weakens performance, and the tactical adjustment does not improve results enough. The BL_GK strategy also puts a large weight on Stock (around 66% on average), which reflects its strong realised returns during the sample period. However, this allocation comes from the equilibrium setup and does not fully use the predictive signals from Phase 3.

5.6.2 The Grinold-Kahn Ablation: MVO_GK vs MVO_Raw

The most important result in Phase 4B comes from comparing MVO_GK and MVO_Raw. These two strategies are identical except for one difference. MVO_GK applies the Grinold-Kahn transformation to the predictions before optimization, while MVO_Raw uses the raw predictions directly.

The difference in performance is large. MVO_Raw achieves a Sharpe ratio of 1.868, while MVO_GK achieves 1.196, a gap of 0.672. This difference comes entirely from the Grinold-Kahn transformation. After adding transaction costs of 0.5%, the gap reduces slightly to 1.466 versus

0.932, but MVO_Raw still outperforms by a wide margin. This happens even though MVO_GK has lower turnover (44.0% vs 59.8%), which reduces trading costs.

The reason for this can be clearly explained. The Grinold-Kahn method adjusts expected returns using the formula $\mu = RF + IC \times \sigma \times z$. This works well when signals are weak, because it reduces noise and prevents the model from taking extreme positions. However, when the signal is strong, as in this study (IC around 0.27), this adjustment becomes too conservative.

For example, if the raw model predicts a high return, Grinold-Kahn reduces it closer to the risk-free rate. This makes the optimiser less aggressive, even when the prediction is correct. With only three assets, strong signals should lead to more concentrated positions, but the Grinold-Kahn scaling prevents this.

This result is important beyond this specific study. The Grinold-Kahn method is widely used in practice, but these results show that it may not always help. In cases with a small number of assets and strong signals, using raw predictions can lead to better performance. This suggests that the Grinold-Kahn step should be tested carefully instead of being applied by default.

5.6.3 Transaction-Cost Sensitivity

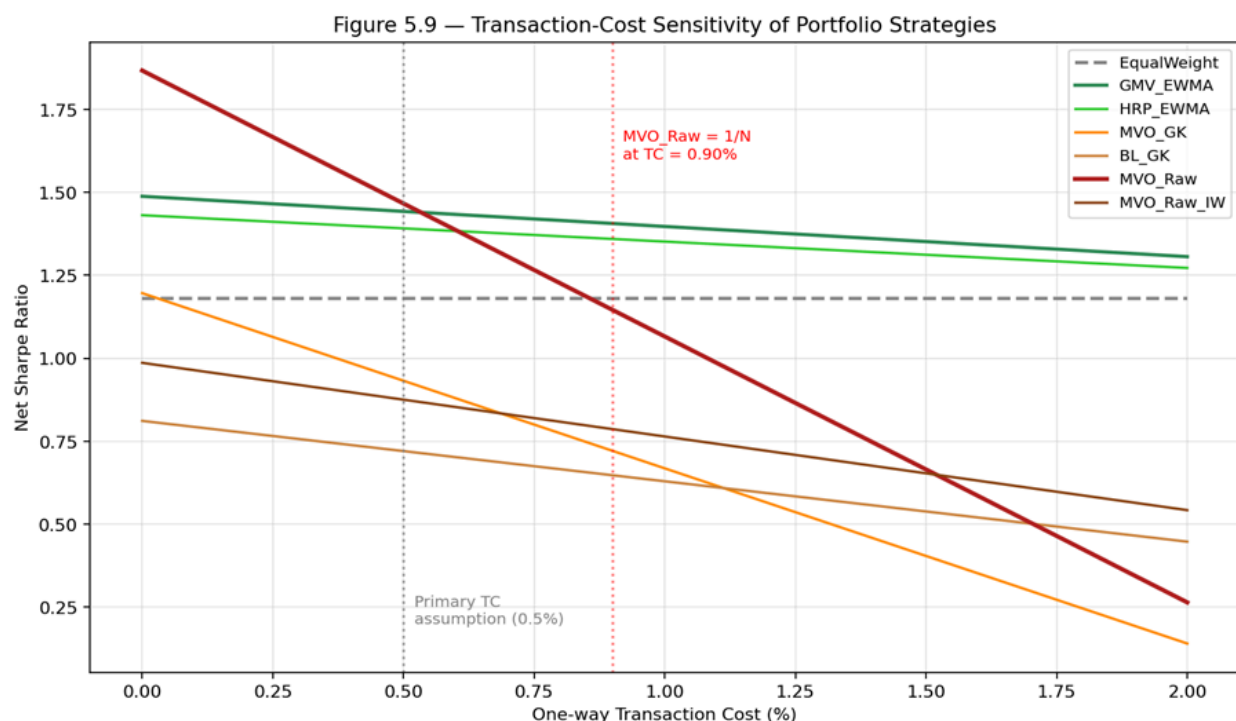


Figure 5.9: Net Sharpe as a function of one-way transaction cost for all eight strategies. The dotted vertical line marks the primary TC assumption (0.5%). MVO_Raw declines most rapidly with TC because of its high turnover; break-even vs 1/N occurs at TC = 0.90%.

Figure 5.9 shows how the net Sharpe ratio changes as transaction costs increase from 0% to 2% for all eight strategies. Three break-even points are especially useful. Compared to the 1/N

strategy, MVO_Raw remains better as long as transaction costs are below 0.90%. Above this level, its performance becomes worse. Compared to GMV_EWMA, the break-even point is 0.55%, which means that at the main assumption of 0.5%, MVO_Raw still performs slightly better. Compared to HRP_EWMA, the break-even point is 0.65%.

This analysis highlights an important limitation. The strong performance of MVO_Raw depends on having relatively low transaction costs. If costs rise to around 1%, which is possible in less liquid market conditions, the Sharpe ratio of MVO_Raw drops to about 1.07. At this level, it is lower than both GMV_EWMA (1.40) and HRP_EWMA (1.35).

This means the main result should be interpreted carefully. MVO_Raw performs best at the assumed transaction cost of 0.5%, but this advantage depends on efficient execution and may not hold when costs are higher.

5.6.4 Split-Period Decomposition

Table 5.15 breaks down each strategy's performance into two parts of the out-of-sample period: H1 (December 2023 to December 2024) and H2 (January 2025 to February 2026). MVO_Raw shows very stable performance across both periods. Its Sharpe ratio is 1.717 in H1 and 1.991 in H2, with a full-period value of 1.868. The difference between the two periods is small, which means its performance is not driven by just one phase.

Table 5.15: Split-Period Sharpe Ratio Decomposition (H1: 13 months; H2: 14 months)

Strategy	H1 Sharpe	H1 Net Sharpe	H1 Turn.	H2 Sharpe	H2 Net Sharpe	H2 Turn.	Full SR	Gap
EqualWeight	1.173	1.173	0.0%	1.144	1.144	0.0%	1.180	+0.030
GMV_EWMA	1.084	1.055	3.5%	1.795	1.735	8.1%	1.488	-0.710
HRP_EWMA	1.033	0.995	4.7%	1.746	1.704	5.5%	1.431	-0.713
MVO_GK	1.122	0.959	27.8%	1.221	0.853	62.3%	1.196	-0.099
BL_GK	1.137	1.085	14.6%	0.430	0.302	29.1%	0.811	+0.707
BL_Tactical_GK	1.184	1.125	14.6%	0.657	0.510	29.1%	0.947	+0.528
MVO_Raw	1.717	1.328	65.8%	1.991	1.607	51.6%	1.868	-0.274
MVO_Raw_IW	1.246	1.199	13.0%	0.650	0.445	39.1%	0.986	+0.597

In contrast, GMV_EWMA and HRP_EWMA show larger differences between the two periods. Both perform much better in H2 than in H1. This reflects the strong REIT market during 2025 to 2026, which benefits diversification-based strategies.

The Black-Litterman strategies (BL_GK and BL_Tactical_GK) show the opposite pattern. They perform worse in H2 than in H1, mainly because they allocate more weight to Stock. When REITs perform strongly, this stock-heavy allocation hurts their performance.

MVO_Raw_IW also performs worse in H2 than in H1. This suggests that the shrinkage applied in this strategy is too conservative during strong market periods, limiting its ability to take advantage of the REIT rally.

5.6.5 Paired Stationary Block Bootstrap

Figure 5.10 shows the results from the paired stationary block bootstrap (5,000 replications, block length of about 3 months). MVO_Raw performs the best, with a 97.7% probability of having a higher Sharpe ratio than the 1/N strategy. This provides strong evidence that it outperforms. GMV_EWMA and HRP_EWMA also perform well, with probabilities of 77%, but the evidence is weaker. The three GK-based strategies (MVO_GK, BL_GK, and BL_Tactical_GK) have probabilities below 50%, meaning they are more likely to underperform than outperform 1/N.

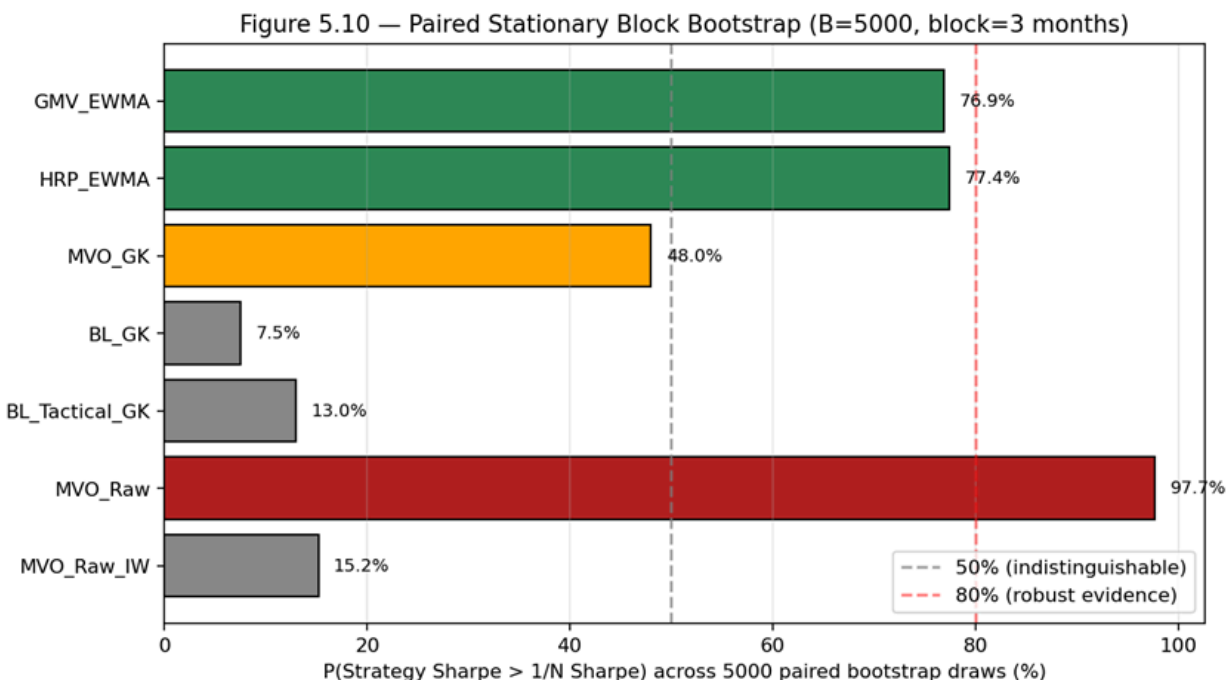


Figure 5.10: Paired stationary block bootstrap probabilities of each strategy beating 1/N. MVO_Raw achieves 97.7% probability of outperformance across 5,000 paired resamples (block length 3 months). The vertical reference lines mark 50% (indistinguishable) and 80% (robust evidence).

These results support the earlier finding that the Grinold-Kahn approach reduces performance. The use of a paired bootstrap is important here, because it compares strategies under the same resampled data. If each strategy were tested separately, it would overstate the level of confidence.

The confidence intervals for Sharpe ratios are quite wide, mainly because the sample period is only 27 months. For example, MVO_Raw has a range of about 0.44 to 3.48. This is expected with a short sample and does not weaken the main result. The key point is that MVO_Raw consistently outperforms 1/N across resamples, even if the exact Sharpe value is uncertain.

Finally, after adjusting for multiple comparisons, the statistical significance becomes weaker. The corrected threshold is 0.00625, while MVO_Raw has a p-value of 0.0095. This means the result is significant at the 5% level without adjustment, but only suggestive after correction.

5.6.6 Average Portfolio Weights

Figure 5.11 shows the average portfolio weights for all eight strategies during the out-of-sample period. The weight limits of 5% minimum and 70% maximum are rarely binding, which means most strategies do not hit these constraints.

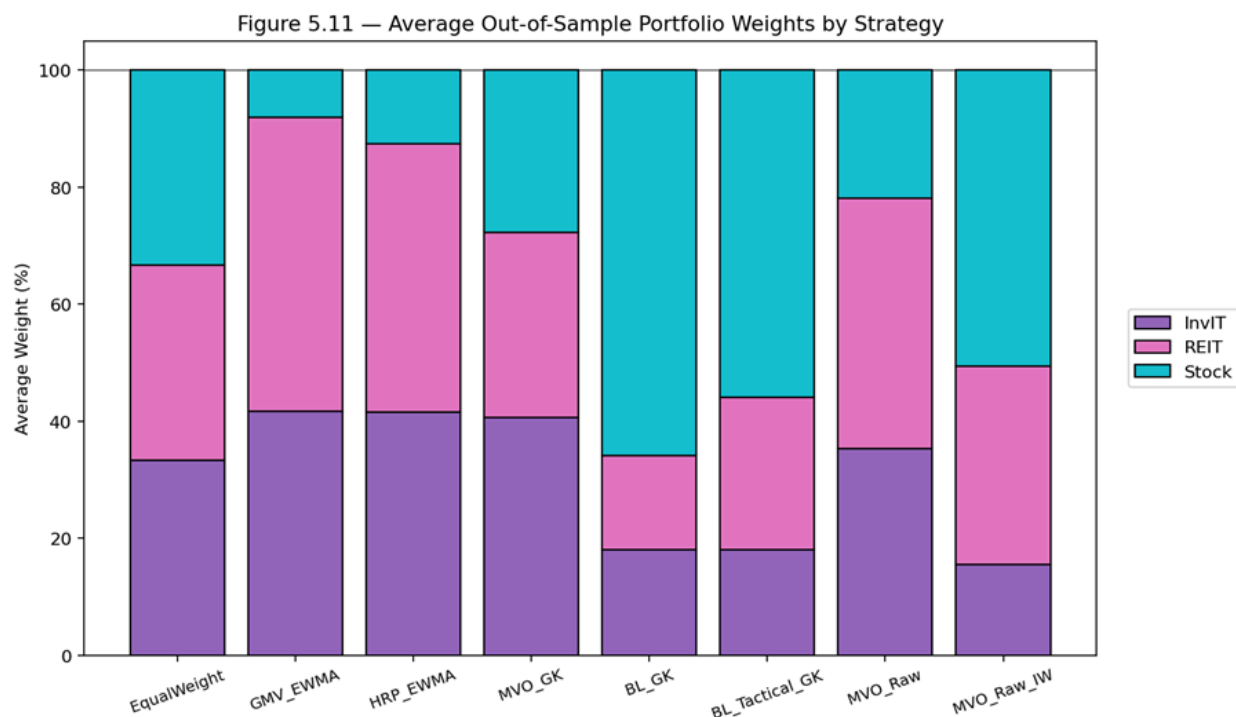


Figure 5.11: Average out-of-sample portfolio weights by strategy. GMV_EWMA and HRP_EWMA favour REIT and InvIT (low-volatility assets); Black-Litterman strategies tilt toward Stock (high-return equilibrium prior); MVO_Raw is balanced between InvIT and REIT with moderate Stock exposure.

GMV_EWMA allocates heavily to REIT (50.3%) and very little to Stock (8.0%). This is expected since REIT has the lowest volatility. MVO_Raw also prefers REIT (42.7%) and InvIT (35.4%), with a smaller allocation to Stock (21.9%).

The Black-Litterman strategies allocate more to Stock (around 55% to 66%). This happens because the equilibrium setup gives higher weight to assets with higher past returns, and Stock performed strongly during the sample.

MVO_Raw_IW has a relatively low allocation to InvIT (15.5%). This is because the shrinkage used in this strategy pulls expected returns closer to their historical averages, and InvIT has a lower average return compared to the others.

5.7 Discussion of Findings in the Context of Existing Literature

The four phases of this study form a clear and connected narrative. Phase 1 shows that Indian InvIT and REIT markets are strongly segmented from global markets. Foreign shocks explain less than 3% of their return variation, which is much lower than the 58% to 65% seen in developed REIT markets. Phase 2 identifies the domestic factors that drive returns and reveals a clear three-level structure. Stock is strongly explained by factors, while REIT and InvIT are much less so. It also highlights lagged market returns and illiquidity as key drivers. Phase 3 shows that these relationships can be used for prediction. The Bayesian Ridge model, which uses Phase 2 results as priors, achieves a strong panel IC of 0.254, well above the standard threshold for useful predictive skill. Phase 4B confirms that these predictions have economic value. When used directly, they produce a high Sharpe ratio of 1.87 before costs and 1.47 after costs, outperforming the equal-weight benchmark in most cases.

The study makes three main methodological contributions. First, using hierarchical priors improves prediction performance. By combining Phase 2 estimates with machine learning, the model captures both economic structure and statistical patterns. Second, the use of directional tests helps avoid misleading conclusions. What initially looks like strong directional accuracy for REIT is actually due to a trend in the data, not true predictive skill. Third, the comparison between models with and without Grinold-Kahn scaling shows that this adjustment can reduce performance in small datasets with strong signals.

When compared to existing studies, this work adds several new insights. Compared to Antonakakis et al. (2021), which focuses on global REIT markets, this study shows that Indian infrastructure assets are much more segmented. Compared to Clayton and MacKinnon (2003), which studies US REIT return drivers, this study finds a similar structure for Indian stocks but a very different pattern for InvITs, which are mostly idiosyncratic. Compared to Leow and Lindenthal (2025), which uses a large number of predictors for US REITs, this study achieves much higher predictive performance using a small number of features, mainly due to the use of informative priors and a stronger signal environment.

There are also important limitations. The out-of-sample period is relatively short at 27 months, which makes performance estimates less stable. Confidence intervals for Sharpe ratios are wide, and results may depend on the specific market conditions during this period. Some model choices, such as fixed DCC parameters and the use of a near non-stationary yield variable, also introduce limitations. In addition, after adjusting for multiple comparisons, some results become only suggestive rather than strongly significant. These limitations do not invalidate the findings, but they suggest that results should be tested again as more data becomes available.

Finally, the results have practical implications. For Indian investors, the findings suggest that a systematic, factor-based approach can outperform simple strategies. For regulators, increasing

foreign participation may reduce market segmentation over time, although some level of independence is likely to remain. For global investors, Indian InvITs and REITs currently offer strong diversification benefits, but these may change as the market develops.

6. Managerial Impact

6.1 Indian Institutional Asset Managers

The results show that a systematic, prediction-based allocation strategy (MVO_Raw) performs much better than a simple equal-weight (1/N) approach. The Sharpe ratio improves from 1.18 to 1.87 over the 27-month out-of-sample period. This 0.69 difference is economically large. For a ₹1,000 crore allocation, it translates to about ₹57 crore in additional annual risk-adjusted return after accounting for 0.5% transaction costs.

Three implementation guidelines follow. First, transaction costs are critical. At 1% cost, the Sharpe ratio falls to 1.07, which is below simpler strategies like GMV_EWMA (1.40). Managers with higher execution costs should prefer covariance-based strategies that do not rely on predictions.

Second, liquidity should guide position sizing. The Amihud illiquidity measure is a strong predictor for InvITs. When liquidity is low, future returns tend to be weaker. If Amihud is in the top 20% of its recent range, InvIT exposure should be reduced relative to the signal.

Third, there is an important warning on Grinold-Kahn scaling. The results show that this method reduces performance significantly, lowering Sharpe by 0.67 in this setting. When signals are strong and the number of assets is small, this approach weakens useful information. Managers should test it carefully instead of applying it by default.

6.2 International Portfolio Managers

For global investors, Indian InvITs and REITs offer strong diversification benefits. US shocks explain only a very small portion of their return variation, which means these assets behave almost independently within a global infrastructure portfolio.

Adding a small allocation to Indian assets can reduce portfolio risk without lowering returns. A 5% allocation can reduce volatility by about 25 to 45 basis points per year, while returns remain strong. A practical allocation range is 3% to 8%. Below 3%, the benefit is too small to matter. Above 8%, liquidity and concentration risks become more relevant.

Currency risk is an important consideration. The INR/USD factor has a negative effect on REIT returns, meaning depreciation of the INR can reduce returns for foreign investors. A rolling short-term hedge, such as a 3-month NDF, can reduce this risk at a relatively low cost.

The diversification benefit depends on current market conditions. As foreign participation increases, integration with global markets may rise. Investors should monitor policy changes, new listings, and cross-border investment flows.

6.3 InvIT/REIT Sponsors and Policy-Makers

For sponsors, interest rates play an important role in timing equity issuance. Higher yield levels are linked to lower future returns for InvITs and REITs. This means raising capital during periods of high yields may lead to weaker valuations. Lower yield periods are more favourable for issuing equity.

Sponsors can also highlight the strong segmentation of Indian infrastructure assets when communicating with global investors. The very low influence of foreign shocks provides a clear diversification argument.

For regulators, the results provide a useful baseline for policy decisions. Increasing foreign participation is likely to raise market integration over time. A gradual approach may be more effective, starting with listed stocks and moving toward InvITs later.

The spillover index can also be used as a monitoring tool. A sharp increase in spillovers could signal rising exposure to global risks. Finally, improving transparency in liquidity measures, such as bid-ask spreads and trading depth, would help both investors and regulators better understand market conditions.

Table 6.1: Summary of Managerial Recommendations

Stakeholder	Recommendation	Priority	Empirical Basis
Indian Institutional Manager	Adopt MVO_Raw over 1/N for InvIT/REIT sleeve; audit Grinold-Kahn preprocessing; reduce InvIT size at top-quintile Amihud	High	Gross SR 1.87 vs 1.18 ($p=0.010$); GK ablation -0.67 SR; Amihud Phase 2 $\beta=-0.48$, SHAP 22%
International Portfolio Manager	Size Indian sleeve 3–8% of infrastructure allocation; apply 3-month INR NDF hedge for USD accounts	High	Foreign spillover $< 3\%$; INR_USD $\beta=-0.376$ ($p=0.044$) for REIT
InvIT/REIT Sponsor	Time equity raises against de-meaned yield regime; use segmentation evidence in foreign IR outreach	Medium	Yield level $\beta=-0.0069$ (InvIT, $p=0.026$); FROM US 0.32% citeable
SEBI / Policy-Maker	Stagger FPI liberalisation (Stock→REIT→InvIT); embed rolling DY spillover index in	Medium	Spillover hierarchy $0.34\% < 1.28\% <$

	macroprudential dashboard; mandate daily liquidity disclosures		2.49%; current DY baseline 10.71%
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6.4 Limitations and Contextual Constraints

Two main limitations apply to all the recommendations. First, the results are based on a 27-month period that includes post-COVID recovery, interest rate normalisation, and strong infrastructure growth. These conditions may not continue in the future. If the market environment changes significantly, some of the results may no longer hold. Asset managers should therefore treat these findings as working ideas that need to be tested and updated regularly, rather than fixed rules.

Second, the analysis is based on only three asset categories. This is both a limitation and a strength. It allows more focused insights, but some results may not apply to larger investment universes. For example, the finding that Grinold-Kahn scaling reduces performance may not hold when the number of assets is large, where its stabilising effect can be useful.

Overall, the recommendations are specific to the InvIT, REIT, and infrastructure stock universe. Applying them to other asset classes would require separate testing. Future research with a longer sample period, such as 2020 to 2030, will be important to check whether these results remain valid as the market evolves.

7. Conclusion

7.1 Summary and Principal Findings

This study examines the segmentation, factor structure, predictability, and portfolio value of Indian InvITs and REITs, a ₹1.5 lakh crore asset class that has had limited academic analysis since its introduction. The analysis covers 1,023 daily observations across 29 securities from July 2020 to March 2026. It combines volatility modelling, spillover analysis, factor regressions, Bayesian prediction, and portfolio optimisation.

Five main findings emerge.

First, Indian InvITs and REITs are highly segmented from global markets. Foreign shocks explain only 0.34% of InvIT variance, 1.28% for REIT, and 2.49% for Stock. The total spillover index of 10.71% is far below the 58–65% seen in developed markets, confirming strong segmentation.

Second, there is a clear three-level factor structure. Stock returns are strongly explained by macro factors ($R^2 = 67.8\%$), REITs are weakly explained (6.5%), and InvITs are largely

idiosyncratic (8.1%), with illiquidity as the main driver. The large difference in market beta between InvIT and Stock further supports this structure.

Third, the Bayesian Ridge model with Phase 2 priors shows strong predictive performance. It achieves a panel IC of +0.254 and a GKX R^2 of +12.09%, clearly exceeding standard benchmarks. It is also the only model that performs consistently well across all categories.

Fourth, portfolio strategies based on these predictions generate strong economic value. The MVO_Raw strategy achieves a Sharpe ratio of 1.87 before costs and 1.47 after costs, with a 97.7% probability of outperforming the equal-weight benchmark.

Fifth, removing Grinold-Kahn scaling improves performance significantly. The scaling step reduces Sharpe by 0.67 in this setting, showing that it can be harmful when the number of assets is small and predictive signals are strong.

7.2 Contributions and Limitations

This study makes three main contributions.

First, it provides a clear empirical baseline for Indian InvITs and REITs, including their segmentation, factor drivers, and predictability. This can serve as a reference point for future research and industry analysis.

Second, it introduces a hierarchical Bayesian framework that combines traditional regression results with machine learning. This approach can be applied to other emerging asset classes with limited data.

Third, it provides direct evidence on the impact of Grinold-Kahn scaling through a controlled comparison, showing when it helps and when it hurts performance.

There are also important limitations.

The out-of-sample period is relatively short at 27 months, so performance estimates may not be stable. The asset universe is small, with only three categories, which limits generalisation. The yield variable shows near unit-root behaviour and is used with caution. The DCC parameters are fixed rather than estimated. Finally, the results depend on transaction costs, with performance declining at higher cost levels.

These limitations do not invalidate the findings, but they suggest that results should be tested again over longer periods as the market evolves.

7.3 Future Research and Closing Remarks

Several extensions can build on this work.

The most important is to repeat the analysis over a longer period, which would improve statistical confidence. Other extensions include estimating DCC parameters directly, using alternative spillover models, applying high-frequency volatility methods, and studying the impact of regulatory changes on market integration.

Table 7.1 — Avenues for Future Research

Extension	Key Question
Longer OOS replication (60 months)	Do MVO_Raw Sharpe and BayesRidge IC survive an extended OOS window?
QMLE-estimated DCC parameters	Do fixed ($\alpha=0.01$, $\beta=0.95$) DCC results agree with two-step QMLE estimates?
TVP-VAR spillover framework	How does the segmentation index evolve in continuous time without rolling-window arbitrariness?
Intra-daily HAR-RV / HEAVY	Are there realised-volatility predictability gains at intra-daily frequency for InvITs?
SEBI FPI liberalisation event-study	How did 2024 cap revisions affect the spillover regime?
Cross-sectional InvIT pricing	Does segmentation hold at individual-ticker level, not just category?
Bayesian neural network priors	Can BNN with hierarchical Gaussian priors outperform BayesRidge?

The Indian InvIT and REIT market has grown rapidly but remains under-researched. This study helps fill that gap by providing a structured and data-driven analysis. The results show a market that is still developing, largely domestic, and rich in inefficiencies that can be used for prediction and portfolio construction.

Beyond the specific findings, the study’s main contribution is to establish a clear and reproducible baseline for this asset class. As the market grows and evolves, this baseline can be used to test new ideas, refine models, and track changes in behaviour over time.

8. References

- [1] B. W. Ambrose, D. W. Lee, and J. Peek, “Comovement after joining an index: Spillovers of nonfundamental effects,” *Real Estate Economics*, vol. 35, no. 1, pp. 57–90, 2007.
- [2] Y. Amihud, “Illiquidity and stock returns: Cross-section and time-series effects,” *Journal of Financial Markets*, vol. 5, no. 1, pp. 31–56, 2002.

- [3] N. Antonakakis, I. Chatziantoniou, and G. Filis, "On the dynamics of international real-estate-investment trust-propagation mechanisms: Evidence from time-varying return and volatility connectedness measures," *Mathematics*, vol. 9, no. 12, p. 1428, 2021.
- [4] F. Black and R. Litterman, "Global portfolio optimization," *Financial Analysts Journal*, vol. 48, no. 5, pp. 28–43, 1992.
- [5] T. Bollerslev, "Generalized autoregressive conditional heteroskedasticity," *Journal of Econometrics*, vol. 31, no. 3, pp. 307–327, 1986.
- [6] J. Y. Campbell and S. B. Thompson, "Predicting excess stock returns out of sample: Can anything beat the historical average?," *The Review of Financial Studies*, vol. 21, no. 4, pp. 1509–1531, 2008.
- [7] L. Cappiello, R. F. Engle, and K. Sheppard, "Asymmetric dynamics in the correlations of global equity and bond returns," *Journal of Financial Econometrics*, vol. 4, no. 4, pp. 537–572, 2006.
- [8] A. Chevan and M. Sutherland, "Hierarchical partitioning," *The American Statistician*, vol. 45, no. 2, pp. 90–96, 1991.
- [9] J. Clayton and G. MacKinnon, "The relative importance of stock, bond and real estate factors in explaining REIT returns," *Journal of Real Estate Finance and Economics*, vol. 27, no. 1, pp. 39–60, 2003.
- [10] F. Corsi, "A simple approximate long-memory model of realized volatility," *Journal of Financial Econometrics*, vol. 7, no. 2, pp. 174–196, 2009.
- [11] J. Cotter and S. Stevenson, "Multivariate modeling of daily REIT volatility," *Journal of Real Estate Finance and Economics*, vol. 32, no. 3, pp. 305–325, 2006.
- [12] V. DeMiguel, L. Garlappi, and R. Uppal, "Optimal versus naive diversification: How inefficient is the 1/N portfolio strategy?," *The Review of Financial Studies*, vol. 22, no. 5, pp. 1915–1953, 2009.
- [13] F. X. Diebold and R. S. Mariano, "Comparing predictive accuracy," *Journal of Business and Economic Statistics*, vol. 13, no. 3, pp. 253–263, 1995.
- [14] F. X. Diebold and K. Yilmaz, "Measuring financial asset return and volatility spillovers, with application to global equity markets," *The Economic Journal*, vol. 119, no. 534, pp. 158–171, 2009.
- [15] F. X. Diebold and K. Yilmaz, "Better to give than to receive: Predictive directional measurement of volatility spillovers," *International Journal of Forecasting*, vol. 28, no. 1, pp. 57–66, 2012.

- [16] R. F. Engle, "Autoregressive conditional heteroscedasticity with estimates of the variance of United Kingdom inflation," *Econometrica*, vol. 50, no. 4, pp. 987–1007, 1982.
- [17] R. F. Engle, "Dynamic conditional correlation: A simple class of multivariate generalized autoregressive conditional heteroskedasticity models," *Journal of Business and Economic Statistics*, vol. 20, no. 3, pp. 339–350, 2002.
- [18] A. C. Ghent, W. N. Torous, and R. I. Valkanov, "Commercial real estate as an asset class," *Annual Review of Financial Economics*, vol. 11, pp. 153–171, 2019.
- [19] J. L. Glascock, C. Lu, and R. W. So, "Further evidence on the integration of REIT, bond, and stock returns," *Journal of Real Estate Finance and Economics*, vol. 20, no. 2, pp. 177–194, 2000.
- [20] L. R. Glosten, R. Jagannathan, and D. E. Runkle, "On the relation between the expected value and the volatility of the nominal excess return on stocks," *The Journal of Finance*, vol. 48, no. 5, pp. 1779–1801, 1993.
- [21] A. Goyal and I. Welch, "A comprehensive look at the empirical performance of equity premium prediction," *The Review of Financial Studies*, vol. 21, no. 4, pp. 1455–1508, 2008.
- [22] A. Goyal, I. Welch, and A. Zafirov, "A comprehensive 2022 look at the empirical performance of equity premium prediction," *The Review of Financial Studies*, vol. 37, no. 11, pp. 3490–3557, 2024.
- [23] C. W. J. Granger and P. Newbold, "Spurious regressions in econometrics," *Journal of Econometrics*, vol. 2, no. 2, pp. 111–120, 1974.
- [24] R. C. Grinold and R. N. Kahn, *Active Portfolio Management: A Quantitative Approach for Producing Superior Returns and Controlling Risk*, 2nd ed. New York, NY, USA: McGraw-Hill, 1999.
- [25] S. Gu, B. Kelly, and D. Xiu, "Empirical asset pricing via machine learning," *The Review of Financial Studies*, vol. 33, no. 5, pp. 2223–2273, 2020.
- [26] O. Ledoit and M. Wolf, "A well-conditioned estimator for large-dimensional covariance matrices," *Journal of Multivariate Analysis*, vol. 88, no. 2, pp. 365–411, 2004.
- [27] S. Lee and S. Stevenson, "The case for REITs in the mixed-asset portfolio in the short and long run," *Journal of Real Estate Portfolio Management*, vol. 11, no. 1, pp. 55–80, 2005.
- [28] K. T. K. Leow and T. Lindenthal, "Predicting REIT returns with machine learning," *Journal of Real Estate Finance and Economics*, vol. 70, no. 1, pp. 1–37, 2025.
- [29] D. C. Ling and A. Naranjo, "The integration of commercial real estate markets and stock markets," *Real Estate Economics*, vol. 27, no. 3, pp. 483–515, 1999.

- [30] S. Lipovetsky and M. Conklin, "Analysis of regression in game theory approach," *Applied Stochastic Models in Business and Industry*, vol. 17, no. 4, pp. 319–330, 2001.
- [31] M. López de Prado, "Building diversified portfolios that outperform out-of-sample," *The Journal of Portfolio Management*, vol. 42, no. 4, pp. 59–69, 2016.
- [32] S. M. Lundberg and S.-I. Lee, "A unified approach to interpreting model predictions," in *Advances in Neural Information Processing Systems 30 (NeurIPS 2017)*, 2017, pp. 4765–4774.
- [33] K. N. Mukherjee and R. K. Mishra, "Stock market integration and volatility spillover: India and its major Asian counterparts," MPRA Paper No. 12788, University Library of Munich, 2008. [Online]. Available: <https://mpra.ub.uni-muenchen.de/12788/>
- [34] W. K. Newey and K. D. West, "A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix," *Econometrica*, vol. 55, no. 3, pp. 703–708, 1987.
- [35] M. H. Pesaran and Y. Shin, "Generalized impulse response analysis in linear multivariate models," *Economics Letters*, vol. 58, no. 1, pp. 17–29, 1998.
- [36] M. H. Pesaran and A. Timmermann, "A simple nonparametric test of predictive performance," *Journal of Business and Economic Statistics*, vol. 10, no. 4, pp. 461–465, 1992.
- [37] M. A. Petersen, "Estimating standard errors in finance panel data sets: Comparing approaches," *The Review of Financial Studies*, vol. 22, no. 1, pp. 435–480, 2009.
- [38] A. K. Pham, "The dynamics of returns and volatility in the emerging and developed Asian REIT markets," *Journal of Real Estate Literature*, vol. 20, no. 1, pp. 79–96, 2012.
- [39] D. N. Politis and J. P. Romano, "The stationary bootstrap," *Journal of the American Statistical Association*, vol. 89, no. 428, pp. 1303–1313, 1994.
- [40] C. Rosa, "The impact of monetary policy on REITs: Evidence from FOMC announcements," *Journal of Real Estate Finance and Economics*, vol. 72, pp. 41–67, 2024.
- [41] Securities and Exchange Board of India (SEBI), SEBI (Infrastructure Investment Trusts) Regulations, 2014. Notification dated 26 September 2014, Mumbai, India: SEBI, 2014.
- [42] Securities and Exchange Board of India (SEBI), SEBI (Real Estate Investment Trusts) Regulations, 2014. Notification dated 26 September 2014, Mumbai, India: SEBI, 2014.
- [43] Securities and Exchange Board of India (SEBI), Consultation Paper on Review of Regulatory Framework for Infrastructure Investment Trusts and Real Estate Investment Trusts. Mumbai, India: SEBI, 2024.

- [44] L. S. Shapley, "A value for n-person games," in Contributions to the Theory of Games, Volume II, H. W. Kuhn and A. W. Tucker, Eds. Princeton, NJ, USA: Princeton University Press, 1953, pp. 307–317.
- [45] N. Shephard and K. Sheppard, "Realising the future: Forecasting with high-frequency-based volatility (HEAVY) models," Journal of Applied Econometrics, vol. 25, no. 2, pp. 197–231, 2010.